

Designing Reservation Reform Without Mass Unrest: An Agent-Based Model of Protest Against an Income-Only Quota in India

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github.com/utsavbalhara/reservation-reform-protest-model

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Abstract

A recurring proposal in India is to replace caste-based reservation with a single income test at ₹8 lakh. We ask how much street protest such a reform would provoke and which designs would keep it small, and build the evidence needed to answer as far as data allow. An eligibility accounting under the actual creamy-layer and EWS rules finds that 6.8% of Indians would lose eligibility and 3.1% would gain it. An allocation model built on India's over-and-above choice rule and five years of JEE (Advanced) rank lists finds that a merged income pool would cost below-line SC candidates –72% of their IIT seats although they keep eligibility. An agent-based model of 640 Census districts, history-matched to protest event counts from four episodes, turns these losses and the symbolic threat to caste recognition into protest. The event data pin down the relative size of past shocks far better than turnout, and frozen out-of-sample predictions of deaths and geography are weak, so the reform's protest level is not identified: it spans more than an order of magnitude across the assumptions the data leave open. The comparison of designs varies much less. Keeping caste quotas with an income filter inside them changes peak turnout by –97% and is the strongest lever in every structural variant and assumption we test, and with probability 95% under priors on intervention effects. Compensating direct losers and expanding seats on the EWS model are the weakest. Whether a consensus commission or grandfathering is the better end-state-preserving lever is not settled, and suppression costs lives whether or not it backfires.

Keywords: reservation policy; affirmative action; protest; agent-based modelling; history matching; event data; matching markets; India

1 Introduction

Few Indian policies generate as much contention as reservation. Quotas for Scheduled Castes (SC), Scheduled Tribes (ST) and Other Backward Classes (OBC) are defended as constitutional remedies for historical exclusion and criticized as caste-based rather than need-based. A proposal that surfaces repeatedly in public debate is to replace caste categories with a single income criterion. Its supporters present it as fairer; its opponents present it as the end of caste-based recognition.

This paper does not take a position on whether that reform is desirable. It asks two narrower questions:

- Q1.** How much street protest would an income-only reform provoke, and from whom?
- Q2.** Which choices of policy design, sequencing and political process would keep that protest small, and which commonly used responses would make it worse?

These questions are hard to answer from data alone. The reform has never been attempted, protest turnout in India is poorly measured, and the mechanisms involved interact non-linearly: loss aversion, group identity, contagion between neighbours, organization and repression. An

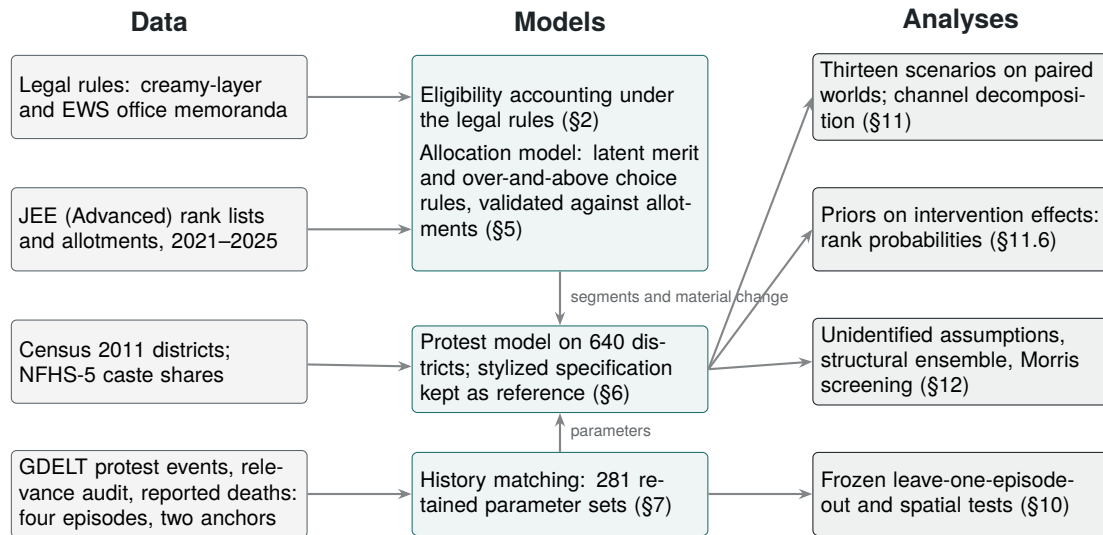


Figure 1: How the analysis fits together. Each model takes its inputs from data where data exist; the analyses on the right separate what the data identify from what is assumed.

agent-based model makes the assumed mechanisms explicit and lets alternative designs be compared under identical conditions (Epstein, 2002; Granovetter, 1978). It also makes it easy to mistake assumptions for findings. Much of this paper separates the two: which inputs can be grounded in data, which conclusions follow from the dynamics rather than from how an intervention was coded, and which survive changes to the model’s structure.

Scope. The project began as a stylized agent-based model of protest, calibrated to a single assumed turnout figure. Answering Q1 and Q2 required three things that model could not supply: who loses under the reform’s legal rules, how a merged pool would reallocate seats, and what past episodes say about the size of shocks. The analysis therefore now links four components (Figure 1): an eligibility accounting under the legal rules; a seat-allocation model built on India’s choice rules and published rank lists; a protest model on a district-level population whose parameters are history-matched to event data from four episodes; and a layer of uncertainty analysis that separates what the data identify from what is assumed. The stylized model is kept as a reference specification, so every result can be compared between an assumption-driven and a data-driven version of the same mechanisms.

Contributions. To our knowledge, and after the searches documented in Appendix A, this is the first agent-based model of reservation politics and the first study to compare reservation reform designs by the protest they would provoke. The mechanisms are not new. The paper contributes:

1. **An allocation accounting of the reform.** Using the over-and-above choice rule that India uses to implement vertical reservation (Aygün and Turhan, 2023b; Sönmez and Yenmez, 2022), a latent merit model fitted to five years of JEE (Advanced) rank lists and validated against published allotments, and the actual eligibility rules for the OBC creamy layer and the EWS quota, we estimate how an income-only pool would redistribute IIT seats, and not only eligibility. The reform reaches beyond SC and ST families above the line. 6.8% of Indians lose eligibility, 3.1% gain it, and below-line SC and ST candidates, who keep eligibility, lose most of their seats to below-line General candidates.
2. **A protest model confronted with event data from several episodes.** A district-level synthetic population (640 Census districts) is history-matched (Andrianakis et al., 2015;

Vernon et al., 2010) to machine-coded protest event counts (GDELT) for four reservation-related episodes, including a null episode and an upper-caste counter-mobilization, with an audited relevance sample, reported deaths, and two observation-model anchors. Leave-one-episode-out predictions were published, with their hashes, before they were scored.

3. **A comparison of interventions that separates inputs from findings.** Thirteen scenarios run on paired worlds with bootstrap intervals. Each lever is decomposed into its channels; each lever's effect size gets a prior, from which we report the probability that each lever is strongest; and the comparison is rerun across the assumptions the data leave open, an eleven-variant structural ensemble and a Morris screening.
4. **A transparent development record**, including two specifications that were badly wrong and how a historical validation test caught them.
5. **An open, reproducible platform.** Code, derived data and results are public; every number in the paper is generated from result files; a test suite includes an independent reimplementation. The allocation model, the scenario catalogue and the calibration machinery can be reused for other reservation designs.

Section 2 sets out the legal architecture, the eligibility accounting and the mobilization record. Section 3 reviews related work and Section 4 the theory. Section 5 presents the allocation model, Section 6 the protest model, and Section 7 the episode data and calibration. Section 8 records the model's development. Sections 9–12 present the experiments, results and robustness analyses. Section 13 discusses what the evidence changed and what it means, Section 14 sets out the limitations, and Section 15 concludes.

2 Background

2.1 The legal architecture of reservation

The Constitution permits special provisions for socially and educationally backward classes and for SCs and STs (Articles 15(4) and 16(4)), and reserves legislative seats for SCs and STs (Articles 330 and 332). Central quotas stand at 15% for SC, 7.5% for ST and 27% for OBC (Deshpande, 2013; Galanter, 1984).

Income tests entered the system through the courts. In *Indra Sawhney v. Union of India* (1992), the Supreme Court upheld the 27% OBC quota introduced after the Mandal Commission, but excluded the “creamy layer” of advanced OBC families and capped total reservation at 50%. The creamy-layer exclusion was reaffirmed for OBC admissions to central educational institutions in *Ashoka Kumar Thakur v. Union of India* (2008). The Constitution (One Hundred and Third Amendment) Act, 2019 added a 10% quota for Economically Weaker Sections (EWS) of the General category. The Supreme Court upheld it by 3:2 in *Janhit Abhiyan v. Union of India* (2022). To introduce EWS without reducing other categories, the government told the Court it had approved 2.14 lakh additional seats in central institutions, about a 25% increase in intake (The Tribune, 2022).

The two income tests are not the same test, and neither is a simple cut at ₹8 lakh of family income.

- **OBC creamy layer.** The income-and-wealth test uses a ₹8 lakh threshold, but income from salaries and from agricultural land is not counted (DoPT, 1993). Separate status criteria exclude, for example, children of senior (Group A) officers. An OBC family whose gross income exceeds ₹8 lakh mainly through salary or farm income is therefore non-creamy and eligible today.

- **EWS.** Eligibility requires family income from all sources below ₹8 lakh and excludes families owning 5 acres or more of agricultural land, a residential flat of 1,000 square feet or more, or residential plots above set sizes (DoPT, 2019). Some General-category families below the income line are therefore ineligible today.

SC and ST quotas remain without any income test. In *State of Punjab v. Davinder Singh* (2024), a seven-judge bench permitted sub-classification within SC and ST quotas, and four of the seven judges said that a creamy-layer principle should also apply to them. The Union government stated it would not apply a creamy layer to SCs and STs. Census 2027 will enumerate caste for the first time since 1931 (PIB, 2025).

2.2 What an income-only test changes: eligibility

We take the reform to be one test on gross family income from all sources at ₹8 lakh, with no category and no asset test. Table 1 sets out who changes status. Two shares are not observed: the share of above-line OBC families who are non-creamy today (prior 0.3–0.7, central 0.5), and the share of below-line General families who fail the EWS asset tests (prior 0.05–0.25, central 0.15). With the central values, 6.8% of the population loses eligibility (range over the priors 5.2%–8.3%) and 3.1% gains it (1.0%–5.1%). Counting only SC and ST families above the line, which is what treating every income test as the same gross-income cut implies, gives 2.9%. The difference is the non-creamy OBC families above the line, who are eligible today and would not be.

Table 1: Eligibility before and after a single gross-income test at ₹8 lakh (district-based synthetic population; central values of the two unobserved shares).

Segment	Share of India	Today	After reform
SC and ST families above ₹8 lakh	2.9%	Reserved	Unreserved
OBC above ₹8 lakh, non-creamy today	3.9%	Reserved	Unreserved
SC and ST families below ₹8 lakh	22.4%	Reserved (own quota)	Reserved (merged pool)
OBC families below ₹8 lakh	38.0%	Reserved (own quota)	Reserved (merged pool)
General below ₹8 lakh, EWS-eligible	17.2%	Reserved (EWS)	Reserved (merged pool)
General below ₹8 lakh, failing EWS asset tests	3.1%	Unreserved	Reserved (merged pool)
OBC creamy layer and General above ₹8 lakh	12.6%	Unreserved	Unreserved

2.3 What an income-only test changes: seats

Eligibility is only half of the material change. Today the reserved categories hold separate quotas. Under the reform the SC, ST, OBC and EWS quotas merge into one pool for all below-line families, so below-line SC and ST candidates compete with below-line General and OBC candidates for the same seats. Whether they gain or lose depends on how the pool is sized and on where each group sits in the merit distribution, which differs by category. Evidence on caste and income as targeting criteria points the same way: caste-based targeting in engineering admissions reaches students poorer than the upper-caste applicants they displace (Bertrand et al., 2010), and caste remains a marker of disadvantage among poor households (Deshpande and Ramachandran, 2019a). Section 5 estimates the seat changes directly.

2.4 Reservation mobilization in India

Reservation has repeatedly produced large mobilizations. In 1990, implementation of the Mandal Commission's OBC quota provoked sustained student protests. In 2006, OBC quotas in central higher education provoked the Youth for Equality movement. Demands for backward-class status drove the Patidar agitation in Gujarat (2015), the Jat agitation in Haryana (2016), the Kapu agitation in Andhra Pradesh (2016), the Gujar agitation in Rajasthan (2019) and the Maratha Kranti Morcha in Maharashtra, whose largest march, in Mumbai on 9 August 2017, drew estimates of 3 to 5 lakh participants.

Four episodes concern SC and ST protections or the income test directly and are used to calibrate the model (Section 7):

- **2 April 2018.** In *Subhash Kashinath Mahajan v. State of Maharashtra* (March 2018), the Supreme Court diluted arrest provisions of the SC/ST (Prevention of Atrocities) Act. No one lost a quota. A Bharat Bandh followed on 2 April; 11 to 14 people died. It was organized with little central leadership or party backing: the first call came from a local leader in Punjab and spread through WhatsApp and local Dalit organizations (Daniyal, 2018). Same-day reports spoke of “thousands” of protesters (Al Jazeera, 2018); later commentary of “hundreds of thousands” (Shepherd, 2018). Parliament restored the provisions later that year.
- **6 September 2018.** Upper-caste organizations called a Bharat Bandh against the restoring amendment. It was strongest in Madhya Pradesh, Bihar and Uttar Pradesh, with no deaths reported.
- **21 August 2024.** After *Davinder Singh*, a Bharat Bandh demanded that no creamy layer or sub-classification apply to SC and ST quotas. Opposition parties including the RJD, the BSP and the Left backed it, yet it drew a mixed response, and SC and ST communities that expected to gain from sub-classification opposed it.
- **January 2019.** The EWS amendment passed with little street protest, which makes it a useful null episode.

From these episodes we draw stylized facts that the model should reproduce or respect:

- S1. A purely symbolic threat to SC/ST protections can produce national, bandh-scale mobilization (2018).
- S2. Mobilization is organized around bandh calls and concentrated on those days.
- S3. Large mobilizations do not require party backing (2018 was locally organized), and party backing does not guarantee them (2024). Party backing is at most one amplifier among several.
- S4. Groups split when a reform creates winners inside them (2024).
- S5. Deaths occur at some large mobilizations and not others, and states often concede afterwards (2018).

Because of S3, the model does not treat opposition-party backing as the source of large mobilizations; the calibration below treats party backing as an episode-specific input.

3 Related work

Reservation design. The closest formal work is in matching theory. Sönmez and Yenmez (2022) formalize how India implements vertical and horizontal reservations, and Aygün and Turhan (2020, 2023a) study the de-reservation of unfilled OBC seats in IIT admissions; Aygün and Turhan (2023b) characterize the over-and-above rule axiomatically. We use their choice

rules rather than inventing an allocation mechanism. Empirical work compares caste and income as targeting criteria (Bertrand et al., 2010; Deshpande and Ramachandran, 2019a,b). An incentivized experiment finds that SC and ST beneficiaries are judged less competent even when affirmative action is income-based (Deshpande et al., 2026), which bears on whether an income test would remove the symbolic charge of reservation. None of this work models the political response to reform.

Protest models. Threshold models (Granovetter, 1978; Schelling, 1978), critical-mass theory (Marwell and Oliver, 1993) and preference falsification (Kuran, 1991) explain why collective action can be all-or-nothing and hard to predict. Epstein (2002) began a line of agent-based models of civil violence; Srbljinović et al. (2003) model ethnic mobilization, and Lemos (2018) compares model output with the size, duration and recurrence of events in the Social Conflict Analysis Database. Thron and Jackson (2015) caution that such models may display generic self-organized criticality rather than the mechanisms their authors intend. That critique applies here, and it is one reason we test mechanisms against several episodes, report which results are inputs, and use a structural ensemble. For India, Wilkinson (2004) shows that electoral incentives shape whether states prevent or permit communal violence, which motivates the concession and policing mechanisms.

Validating agent-based models. Windrum et al. (2007) survey empirical validation of agent-based models. We follow the history-matching approach of Vernon et al. (2010) and Andrianakis et al. (2015), which rules out parameter sets inconsistent with data rather than fitting a single best set, and which suits a stochastic model with several unidentified shock sizes. Simulation-based inference (Cranmer et al., 2020) is an alternative that we leave for future work. For global sensitivity analysis we use Morris screening (Campolongo et al., 2007; Morris, 1991; Saltelli et al., 2008). Appendix F summarizes the model in the ODD format (Grimm et al., 2020).

Event data. GDELT (Leetaru and Schrodtt, 2013) codes events automatically from news reports. It over-reports events in capitals and media centres (Hammond and Weidmann, 2014), which our observation model allows for. ACLED (Raleigh et al., 2010) covers India from 2016 and would be a natural cross-check; we did not use it here.

4 Theory and hypotheses

4.1 Loss aversion and the politics of retrenchment

Prospect theory holds that losses loom larger than equivalent gains, with a loss-aversion coefficient near 2.25 (Kahneman and Tversky, 1979; Tversky and Kahneman, 1992). Pierson (1994, 1996), building on Weaver (1986), argues that retrenchment is harder than expansion because costs are concentrated, immediate and traceable while benefits are diffuse. Governments that retrench successfully rely on blame-avoidance strategies: *obfuscation*, which lowers the visibility of cuts; *division*, including grandfathering, which spares current beneficiaries; *compensation* for losers; and *blame-sharing* through broad coalitions. We model the transparent versions of these strategies. Obfuscation is excluded on ethical grounds (Section 16).

4.2 Group identity and symbolic threat

Social identity theory holds that part of individuals' self-concept derives from group membership, and that perceived threats to a group's status provoke collective responses (Tajfel and

Turner, 1979). For SC and ST communities, caste-based reservation is a constitutional acknowledgement of historical injustice as well as a material benefit. Removing caste as the criterion is therefore a symbolic threat independent of any individual's eligibility. Stylized fact S1 is direct evidence that this channel operates at national scale; the 2018 bandh followed a ruling that removed no one's quota.

4.3 Thresholds, cascades and organization

In threshold models, each individual joins once the share of others participating exceeds a personal threshold (Granovetter, 1978; Schelling, 1978). Aggregate outcomes then depend on the whole distribution of thresholds, and a small change can decide whether a cascade spreads or stops (Watts, 2002). Critical-mass theory stresses that a small, highly motivated core can make collective action viable for everyone else (Marwell and Oliver, 1993), and preference falsification means that latent support can be invisible until a cascade begins (Kuran, 1991). Resource-mobilization and political-process theories stress that organizations and political opportunities convert latent grievance into action (McAdam, 1982; McCarthy and Zald, 1977; Tarrow, 1994).

4.4 Repression and backfire

The relationship between repression and dissent is contested: repression sometimes deters and sometimes escalates (Davenport, 2007; Lichbach, 1987). Hess and Martin (2006) describe conditions under which repression backfires by generating outrage. For India, Rydzak (2019) finds that internet shutdowns are associated with increases in violent collective action rather than decreases, and Rydzak et al. (2020) find little evidence that shutdowns quell protest elsewhere. Because the literature supports both deterrence and backfire, the model does not assume either: the response of turnout to deaths is a parameter whose prior runs from deterrence to strong backfire.

4.5 Hypotheses

- H1. Grandfathering and a glide path reduce protest by deferring material loss (loss aversion).
- H2. Expanding seats reduces protest, following the EWS 2019 precedent.
- H3. Retaining caste categories with an income filter inside them reduces protest by reducing symbolic threat.
- H4. Sub-classification favouring the most-deprived reduces protest by splitting the coalition (S4).
- H5. A data-first commission with cross-party consensus reduces protest by reducing organization and symbolic threat.
- H6. Compensating direct losers reduces protest (Pierson's compensation strategy).
- H7. Credible statutory guarantees reduce protest by limiting feared future changes.
- H8. Internet shutdowns reduce protest.
- H9. Heavy policing reduces protest.

5 The allocation model: who gains and loses seats

5.1 Choice rules

India implements vertical reservation with the *over-and-above* rule (Aygün and Turhan, 2023b; Sönmez and Yenmez, 2022). Open-category seats go first to the highest-merit applicants of

any category; each reserved category's seats then go to its highest-merit remaining members. Reserved-category candidates who would win an open seat on merit therefore do not use up a reserved seat. We apply the same rule to the reform, with the open share unchanged at 40.5% and one reserve category, "family income below ₹8 lakh", holding the rest (59.5% in the main case; 49.5% and 69.5% as variants).

Actual IIT admission is a centralized multi-programme assignment in which top reserved-category candidates often take reserved seats at preferred programmes. For the aggregate number of seats by category, this makes the outcome closer to a *reserved-first* order, in which reserved seats are filled before open seats. We check both orders against the actual category-wise allotments (Section 5.4) and use the one that reproduces them as the main case. Horizontal reservations and candidates' preferences over programmes are not modelled.

5.2 Latent merit

The JEE (Advanced) Common Rank List records each qualified candidate's rank and category. We transcribed the lists for 2019–2025 (Appendix B) and model each category's candidates as normally distributed on a common latent merit scale, with the General category fixed at mean 0 and standard deviation 1. The number of candidates per category comes from the official reports. Given the parameters, rank r sits at the point z where the expected number of candidates above z equals $r - \frac{1}{2}$, and the probability that the candidate at that rank belongs to category c is proportional to $N_c \phi_c(z)$. The means and standard deviations maximize the likelihood of the observed categories at every rank. The 2019 and 2020 lists failed an internal consistency check and are excluded; the fits use 2021–2025.

5.3 Family income

Family income is not observed per candidate. Each category's share of candidates above the line is drawn from a prior: General 0.70–0.95 (the category includes OBC creamy-layer candidates and General candidates above the EWS line), OBC-NCL 0.10–0.40 (above the line on gross income but non-creamy), SC and ST 0.10–0.45; GEN-EWS candidates are below by definition. A merit gradient (0–0.8 on the logit scale per standard deviation of merit) lets richer candidates sit higher in each category's distribution. A draw is kept only if the implied share of admitted IIT students above the line is 28–45%, the range consistent with the parental-income tables of the JEE reports once income growth is allowed for. We make 300 draws per year and keep 1,500 in total.

5.4 Validation

In sample, the fitted category shares match the observed shares by rank decile to within 3.5 percentage points. Out of sample, the GEN-EWS and OBC-NCL rank lists end at the same aggregate-marks cutoff, as do the SC and ST lists, so the latent cutoffs implied by each list's length should agree within each pair. They agree to within 0.012 standard deviations for GEN-EWS and OBC-NCL, and to within 0.03 for SC and ST in every year except 2023 (0.126). Under the reserved-first order, the model's status-quo seat counts by category match the actual IIT allotments to within 406 seats per category and year; the open-first order misses by up to 2,386 seats. The reserved-first order is therefore the main case. NEET, where the same approach would cover medical admissions, is excluded because the fitted tails failed a check against the top-ranked candidates.

5.5 Results

Table 2 gives the seat changes by segment in the main case. The reform does not only take seats from SC and ST families above the line. Below-line SC candidates lose -72% of their seats and below-line ST candidates -85% . The gains go to below-line General candidates ($+158\%$ for those who fail the EWS asset tests today and $+71\%$ for GEN-EWS) and, modestly, to below-line OBC candidates ($+23\%$). At the same income, SC and ST candidates sit lower on the fitted merit scale, so a pool open to all below-line families admits fewer of them. The direction of every change is the same under both processing orders and all three pool sizes (Appendix C).

Table 2: IIT seats by segment under today’s quotas and under a merged income-only pool (reserved-first order, pool 59.5%). Medians over 1,500 draws of the income overlay across 2021–2025; brackets give 90% intervals. Relative material change is the change in admission chances per candidate in units of the loss of an above-line SC candidate (positive = loss); it is the material input of the grounded protest model.

Segment	Seat change	90% interval	Relative material change	90% interval
SC, above the line	-96%	-98 to -93	1.00	1.00 to 1.00
SC, below the line	-72%	-75 to -68	0.66	0.50 to 0.75
ST, above the line	-98%	-99 to -96	1.09	1.00 to 1.17
ST, below the line	-85%	-88 to -81	0.84	0.62 to 0.96
OBC (non-creamy), above the line	-87%	-92 to -79	0.74	0.64 to 0.84
OBC, below the line	$+23\%$	$+10$ to $+37$	-0.16	-0.27 to -0.08
General, EWS-eligible	$+71\%$	$+56$ to $+87$	-0.45	-0.57 to -0.35
General, below the line, not EWS-eligible	$+158\%$	$+144$ to $+189$	-2.61	-3.13 to -1.84
General (incl. OBC creamy), above the line	-10%	-16 to -7	0.18	0.11 to 0.30

The same model evaluates the two levers that are themselves allocation rules. Expanding seats by a quarter, the scale used to introduce EWS in 2019, softens the below-line SC loss to -63% but does not remove it, because the merged pool still ranks SC candidates against General candidates. Keeping the caste quotas but closing them to above-line SC and ST families, whose seats go to below-line candidates of the same category, reverses the sign for the below-line majority: below-line SC candidates gain $+32\%$ and below-line ST candidates $+37\%$. Brazil’s 2012 quota law (Law No. 12.711) combines criteria in this way, with racial sub-quotas inside an income- and school-based quota.

6 The protest model

The model is summarized in the ODD format (Grimm et al., 2020) in Appendix F; this section gives the details. Figure 2 shows what drives one agent’s decision each day. The model has two specifications. The *stylized* specification is the original model, retained as a reference and for comparison with earlier results. The *grounded* specification replaces its assumed inputs wherever data allow; Table 3 lists the differences.

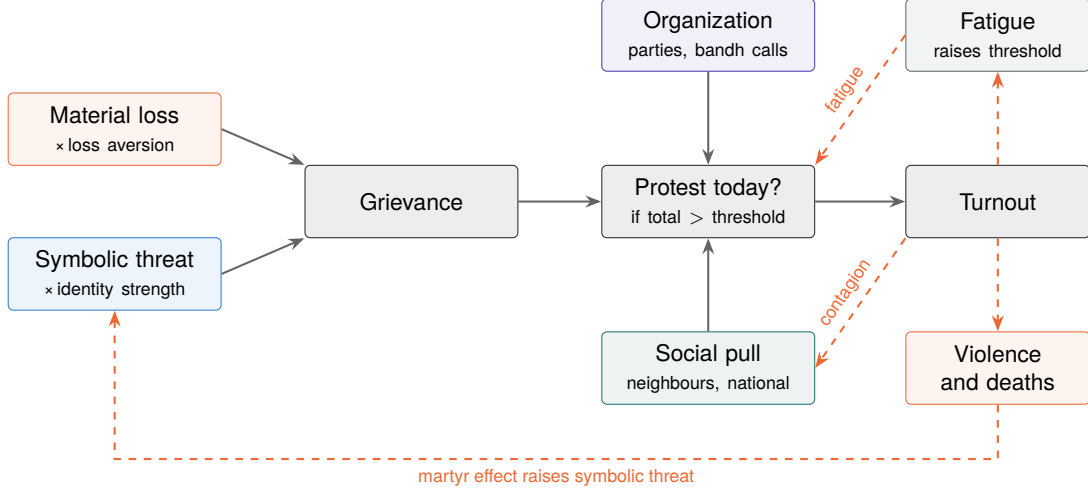


Figure 2: What drives one agent’s decision each day. Dashed arrows are feedback loops: today’s turnout shapes tomorrow’s decisions through contagion, fatigue and the martyr effect.

6.1 Population

The grounded population has $N = 120,000$ agents, each representing about 12,200 people of a population of 146 crore. Agents are allocated to the 640 Census 2011 districts in proportion to population. Within each district the SC and ST shares are the district’s own (Census 2011); the rest are split between OBC and General with the state’s ratio of OBC to OBC-plus-other household heads in NFHS-5. National shares are then SC 16.7%, ST 8.6%, OBC 45.7% and General 29.0%. SC and ST agents are assigned with equal probability to a better-off tier or a most-deprived tier. The probability of being above the ₹8 lakh line is 30% for General and 17% for OBC; for SC it is 19% in the better-off tier and 5% in the most-deprived tier, and for ST 16% and 4%. These shares order groups consistently with survey evidence on the caste gradient of income and wealth (Bharti, 2018; PRICE, 2023). Above-line OBC agents are creamy layer with probability 0.5, and below-line General agents fail the EWS asset tests with probability 0.15.

Each agent draws an identity strength $\iota_i \sim \text{Beta}(2, 3)$ and a standard-normal threshold score z_i . Neighbourhoods of about 100 agents are formed within state and group blocks ordered by district, so a neighbourhood spans about three districts. A share h of agents live in single-group neighbourhoods and the rest in mixed neighbourhoods whose composition matches the national mix; h is estimated in calibration. The stylized population has the same structure without districts, with $h = 1$ (complete segregation), NFHS-4 group shares (OBC 42%, General 32.8%), and no distinction between the creamy layer and the income line.

6.2 Grievance

Agent i in eligibility segment k has a material change m_k (positive for a loss). In the grounded specification m_k is the relative material change of Table 2, which puts the loss of an above-line SC candidate at 1. In the stylized specification it is 1 for SC and ST above the line, 0.35 for SC and ST below, 0.25 for OBC below, -0.15 for General below, and 0 otherwise. Losses are weighted by loss aversion, $L(m) = \lambda m$ for $m > 0$ and m otherwise. Symbolic threat s_g is 1.0 for SC, 0.9 for ST, 0.45 for OBC and -0.4 for General, times 0.8 for the most-deprived tier; in the grounded specification it is scaled by the 2018 episode’s estimated magnitude (Section 7). Grievance on day t is

$$G_{i,t} = w_m L(m_{k(i)}) + w_s \iota_i (\tilde{s}_i + \mu_{g,t} \mathbb{I}[\tilde{s}_i > 0]), \quad (1)$$

where $\tilde{s}_i$ is the agent's tier-adjusted symbolic threat and $\mu_{g,t}$ the accumulated martyr effect for the group.

6.3 Social influence, organization and decision

Let $n_{c,t}$ be the share of neighbourhood c protesting on day t and $v_{g,t}$ the national share of group g . Social pull saturates:

$$S_{i,t} = \beta_n \tanh\left(\frac{n_{c(i),t-1}}{n^*}\right) + \beta_v \tanh\left(\frac{v_{g,t-1}}{v^*}\right). \quad (2)$$

Organization contributes $a o_g b_t$, where o_g is group organizational capacity (SC 0.6, ST 0.4, OBC 0.4, General 0.1), a the opposition-party amplifier and b_t the mobilization level: 1.0 on bandh days and 0.15 otherwise. The agent protests with probability

$$P_{i,t} = \left[1 + \exp\left(-\frac{G_{i,t} + S_{i,t} + a o_g b_t - c_t - \theta_{i,t}}{\tau}\right)\right]^{-1}, \quad \theta_{i,0} = \bar{\theta} + \sigma_\theta z_i, \quad (3)$$

where c_t is a turnout cost imposed by heavy policing on bandh days. Each day of protest raises an agent's threshold by the fatigue increment f .

6.4 Bandh calls, concession and backlash

In the stylized specification bandhs are called on fixed days (3, 13 and 26). In the grounded specification, organizations call a bandh for the next day once the turnout that a bandh would draw, given today's grievance and social pull, reaches 10 lakh. The first call can take effect on day 3; after each bandh there is a refractory period of 10–13 days, and there are at most four calls. The government concedes with a daily hazard that rises logistically with pressure, defined as the largest daily turnout so far in crore plus deaths so far divided by 20 (midpoint 1, width 0.25), up to 15% a day. A concession cuts material loss and the threatened groups' symbolic threat by 60%, cancels any scheduled bandh, and gives the General category a symbolic threat of 0.3, which can produce a counter-mobilization.

6.5 Deaths and the martyr effect

In the stylized specification, deaths on day t are Poisson with mean κ times the day's protester-days in crore, and every death raises μ_g by $\delta = 0.012$ for each group in proportion to its share of protesters, up to 0.35. In the grounded specification deaths are negative binomial with dispersion estimated in calibration, the rate on bandh days is ten times the ordinary-day rate, and deaths are split into police-attributed deaths (half at baseline) and others. A response parameter ρ multiplies the martyr effect: $\rho = 1$ is the reference backfire, $\rho = 0$ no response, and $\rho < 0$ deterrence. Heavy policing raises only police-attributed deaths.

6.6 Uncertainty and paired sampling

Each Monte Carlo run draws one set of uncertain parameters (a world). In the stylized specification this means uniform multipliers on λ (0.7–1.3), symbolic threat (0.8–1.2), a (0.85–1.15), β_n and β_v (0.8–1.2 each) and κ (0.5–2.0), plus $\mathcal{N}(0, 0.1)$ on $\bar{\theta}$. In the grounded specification, the parameters informed by the episodes ($\bar{\theta}$, σ_θ , the reform's symbolic threat, the bandh-day death rate and dispersion) are instead taken from one parameter set drawn at random from those that survive calibration, and the reform's symbolic threat is also multiplied by 0.8–1.2. Run r uses world seed $1000 + r$ and campaign seed $5000 + r$ for every scenario, so scenario comparisons are paired. Effects are reported as the median over runs of the paired ratio (scenario over baseline

in the same world), with 95% percentile-bootstrap intervals over runs (Efron and Tibshirani, 1993), and as the share of runs in which the scenario exceeds the baseline.

Table 3: Stylized and grounded specifications.

Component	Stylized	Grounded
Population	National shares; single-group neighbourhoods	640 districts (Census 2011, NFHS-5); mixing h estimated
Eligibility	One gross-income line for every test	Creamy-layer and EWS asset rules
Material change	Assumed per segment	IIT allocation model (Table 2)
Symbolic threat	Assumed	2018 episode magnitude $\times R$, $R = 1$
θ, σ_θ	Tuned to 20 lakh–1 crore peak	History-matched to four episodes
Bandh calls	Fixed days	Called when expected turnout is credible
Concession	None	Pressure-dependent hazard; General backlash
Deaths	Poisson; all trigger martyr effect	Negative binomial, split by attribution; bandh-day rate estimated
Random streams	One stream	Separate streams for decisions, deaths and events

Table 4 lists every parameter, its value in each specification and where the value comes from.

One campaign. Build the population; draw a world; apply the scenario’s interventions. Then for $t = 0, \dots, 39$:

1. compute $G_{i,t}$, $S_{i,t}$ and $P_{i,t}$ for every agent;
2. draw each agent’s protest decision and raise the thresholds of those who protest;
3. update neighbourhood and national turnout; decide whether a bandh is called for tomorrow and whether the government concedes;
4. draw deaths and update the martyr effect.

Outputs: peak-day protesters, unique participants, protester-days, deaths, bandh days called, concession day, and daily series by group and state.

7 Episode data and calibration

7.1 Event counts

Protest turnout in India is rarely counted, and the estimates that exist differ by an order of magnitude (for 2018, from “thousands” to “hundreds of thousands”). Instead of calibrating to a head-count, we calibrate to the number of protest events reported. From the GDELT 1.0 daily event files we take every event with CAMEO root code 14 (protest) located in India on each episode’s core days, and keep those whose source article URL matches episode-specific keywords (Appendix B). Keyword filters admit irrelevant events: a fuel-price bandh on 10 September 2018, unrelated state bandhs in August 2024, and a general strike during the EWS amendment. For each episode a sample of matched URLs was labelled as relevant or not from the article’s URL text; the posterior precision on core days, $\text{Beta}(1 + \text{relevant}, 1 + \text{irrelevant})$, multiplies the count. To absorb GDELT’s changing coverage, counts are expressed per 1,000 GDELT events located in India on the same days. Table 5 lists the episodes and targets.

Table 4: Parameters. Basis: *L* literature; *D* data (with the source); *H* history-matched to the episodes (median and 90% interval of the retained sets); *T* tuned to the stylized turnout target; *A* assumed; *P* prior range. Parameters marked with an asterisk are varied in the uncertainty analyses of Sections 11.6–12.

Symbol	Role	Stylized	Grounded	Basis
λ	Loss aversion	2.25	2.25	L
w_m^*	Weight of material change	0.45	0.45 (not identified; swept)	T / A
w_s	Weight of symbolic threat	3.3	3.3 (scale absorbed by the shock size)	T
s_g	Symbolic threat, SC/ST/OBC/General	1/0.9/0.45/−0.4	same $\times$ 2018 magnitude 1.59 (0.50–2.38)	A / H
R^*	Reform shock relative to 2018	–	1	A
	Deprived tier’s share of threat	0.8	0.8	A
m_k	Material change by segment	1/0.35/0.25/−0.15	Table 2	A / D (JEE)
o_g	Organizational capacity	0.6/0.4/0.4/0.1	same	A
a^*	Party amplifier	1.5	1.5 (backing 0.6)	A
β_n, n^*	Neighbourhood influence, saturation	0.9, 0.10	same	A
β_v, v^*	National influence, saturation	0.6, 0.02	same	A
$\bar{\theta}$	Mean threshold	6.6	7.06 (5.35–8.35)	T / H
σ_θ	Threshold spread	1.4	1.75 (0.99–2.16)	T / H
τ	Decision noise	0.3	0.3	A
f	Fatigue per protest day	0.10	0.10	A
h	Share of single-group neighbourhoods	1	0.78 (0.54–0.98)	A / H
b_t	Mobilization on bandh / other days	1.0 / 0.15	same	A
	Bandh calls	days 3, 13, 26	expected turnout $\geq$ 10 lakh; gap 10–13 days; ≤ 4	A
κ	Deaths per crore protester-days	4 (all days)	60 on bandh days (5–769); a tenth otherwise	A / H
	Death dispersion	Poisson	negative binomial, H	H
	Police-attributed share of deaths	–	0.5	A
δ, cap	Martyr effect per death, cap	0.012, 0.35	same	A
ρ^*	Response of turnout to deaths	1	1	A (prior −0.5 to 2 for suppression)
	Concession hazard	–	logistic in pressure, $\leq 15\%$ /day	A
	Backlash after concession	–	General threat 0.3	A
	Above-line OBC share non-creamy	–	0.5	P (0.3–0.7)
	Below-line General share failing EWS asset tests	–	0.15	P (0.05–0.25)
	Group shares	Census 2011, NFHS-4	Census 2011, NFHS-5 by district	D
	Above-line shares by group and tier	30/17/19–5/16–4%	same	L (Bharti, 2018; PRICE, 2023)

Table 5: Episodes and calibration targets. Events: keyword-matched GDELT protest events on core days; precision from the audited sample; rate per 1,000 India events on the same days (median and 90% interval). Role: M mechanism episode (simulated), A anchor for the observation model, O held for description only.

Episode	Role	Events	Precision	Rate per 1,000	Deaths	Party backing
SC/ST Bharat Bandh, 2 April 2018	M	308	0.96	31.3 (27.7–32.4)	11–14	0.2
Upper-caste bandh, 6 September 2018	M	77	0.86	7.5 (6.1–8.3)	0	0.1
SC/ST Bharat Bandh, 21 August 2024	M	90	0.85	10.2 (8.2–11.4)	0	0.6
EWS amendment, January 2019	M	247	0.09	1.6 (0.3–4.2)	0	0.0
Maratha march, Mumbai, 9 August 2017	A	56	0.96	6.4 (5.6–6.6)	0	0.5
Patidar agitation, August 2015	A	77	0.86	2.5 (2.0–2.7)	10–11	0.1
Jat agitation, February 2016	O	204	0.96	5.5 (4.9–5.7)	30–31	0.2
Kapu agitation, January 2016	O	9	0.89	1.5 (1.0–1.7)	0	0.1
Maratha quota protests, 2018	O	214	0.96	8.9 (7.9–9.2)	2–8	0.4
Gujjar agitation, February 2019	O	54	0.86	1.1 (0.9–1.2)	0	0.1

7.2 Simulating an episode

Each mechanism episode is simulated as a two-day campaign, an announcement day and a bandh day, with no material change. Its shock is a symbolic threat vector (which groups feel threatened, and in which direction) scaled by an unknown magnitude. For 2018 the vector is (1, 0.9, 0, −0.2) for SC, ST, OBC and General; for the September 2018 upper-caste bandh (−0.3, −0.3, 0.3, 1); for 2024 (1, 0.9, 0, 0) with the most-deprived tier’s share of the threat estimated, since many of them stood to gain from sub-classification; and for the EWS amendment the reform’s own vector. Party backing is coded from reporting: 0.2 for 2018 (locally organized), 0.1 for September 2018, 0.6 for 2024 and 0 for 2019. It sets the party amplifier, with backing 0.6 corresponding to the reform scenario’s $a = 1.5$. The magnitudes are relative to the reform’s symbolic threat to SC agents, so a 2018 magnitude of 1 means that the reform threatens caste recognition as much as the 2018 ruling did.

7.3 Observation model

Expected reported events are $\alpha \sum_s (P_s/10^5)^\gamma$, where P_s is protester-days in state s , with Delhi’s term multiplied by a media factor, since national media concentrate there (Hammond and Weidmann, 2014). The concave form means that a bandh spread over many states produces more reported events than a single rally of the same total size. The scale α , exponent γ and Delhi factor are unknown. They are tied down by two anchor episodes with published crowd estimates: the Maratha march in Mumbai on 9 August 2017 (1 to 20 lakh person-days) and the Patidar agitation of August 2015 (3 to 20 lakh).

7.4 History matching

Fifteen unknowns get uniform priors: $\bar{\theta}$ (5–8.5), σ_θ (0.8–2.2), h (0.5–1), the four episode magnitudes, the 2024 deprived-tier share, the bandh-day death rate ($10^{0.3}$ to 10^3 per crore protester-days), its dispersion, the three observation parameters and the two anchor crowd sizes. For each parameter set, every mechanism episode is simulated with two seeds. A set is *implausible*

if, for any target, the standardized distance

$$I = \frac{|\log \hat{y} - \log y|}{\sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{Poisson}}^2 + \sigma_{\text{sim}}^2 + \sigma_{\text{disc}}^2}} \quad (4)$$

exceeds 3 (Andrianakis et al., 2015; Vernon et al., 2010), where the variances are observation error (from the precision interval), Poisson counting error, simulation noise between seeds, and a model discrepancy of 0.5 on the log scale. Deaths on the bandh day are compared on the natural scale with negative-binomial variance and a discrepancy of 2 deaths. Three waves of 6,000 sets are run; each later wave samples within the range of the previous wave’s survivors. The sets that survive all waves form the not-ruled-out-yet (NROY) set used by the grounded specification.

7.5 Out-of-sample tests

1. **Leave one episode out.** For each of the three bandh episodes, a separate history match uses only the other episodes. Its survivors predict the held-out episode’s bandh-day deaths and its distribution of protest across states; the held-out magnitude is not identified without that episode’s data, so the prediction integrates over its prior. The predictions are written to files and hashed (SHA-256) before a separate script compares them with the observations.
2. **Spatial distribution.** Event counts by state are not used in calibration. For each episode, the Spearman correlation across large states between the model’s protester-days and the observed event counts is compared with two naive baselines: state population and state SC-plus-ST population.

8 Model development: what went wrong and how it was caught

The stylized model is the fourth specification. We report the earlier three because their failures show which mechanisms drive the results and why a model that hits a turnout target can still be wrong. Every number in this section comes from `experiments/replay_model_development.py`, which re-runs each earlier specification with the current code.

Iteration 1: unbounded contagion. The first specification used linear social influence, $S_{i,t} = 1.6 n_{c,t-1} + 25 v_{g,t-1}$, narrow thresholds ($\sigma_\theta = 0.55, \tau = 0.22$) and equal weight on material loss and symbolic threat ($w_m = 1.0, w_s = 2.2$). At every mean threshold tried (2.8 to 3.3), it placed about 98.0 crore people on the street on the peak day, nearly every SC, ST and OBC agent. Even without contagion, the first bandh day would have drawn 30.6% of SC agents, against 0.70% in the final model; and linear contagion has no ceiling, so once a group’s national turnout reached a few percent the visibility term alone exceeded any threshold. The fix was to bound social influence with a saturating function.

Iteration 2: saturating contagion, narrow thresholds. With saturating influence the model could be calibrated: at $\bar{\theta} = 5.1$ it gave a baseline peak of 30.5 lakh. Turnout fell 3.1-fold between $\bar{\theta} = 4.9$ (2.6 crore) and 5.0 (86.6 lakh), so any intervention that lowered grievance slightly appeared to eliminate protest. Grandfathering changed the peak by -99% and compensation by -91% . Removing all material loss, the analogue of the 2018 ruling, left 10,000 protesters, 0.3% of the baseline, which contradicts S1. Iteration 2 matched a plausible turnout but not the documented episode.

Iteration 3: reweighting alone does not fix it. Raising w_s and lowering w_m , recalibrating $\bar{\theta}$ each time, left the symbolic-only shock below 5% of the baseline however the weights were set (Table 6). With a narrow threshold distribution, removing any component of grievance drops the population below the tipping point together.

Table 6: Iteration 3: reweighting symbolic and material grievance with narrow thresholds (6 runs each).

w_s	w_m	$\bar{\theta}$	Baseline peak (lakh)	Symbolic-only peak (lakh)	Symbolic-only share
3.6	0.5	5.3	145.8	2.6	1.8%
5.0	0.4	6.4	76.9	2.7	3.5%
6.0	0.3	7.1	95.8	4.3	4.5%

Iteration 4: a wide threshold distribution. Granovetter’s central point is that the distribution of thresholds, not only its mean, governs collective outcomes. Widening the spread to $\sigma_\theta = 1.4$, with $\tau = 0.3$, made the response smooth. Turnout falls about 2.3-fold per quarter-step of $\bar{\theta}$ near the calibrated value. After recalibration ($w_s = 3.3$, $w_m = 0.45$, $\bar{\theta} = 6.6$), the symbolic-only shock yields 29% of baseline turnout. Figure 3 compares iterations 2 and 4.

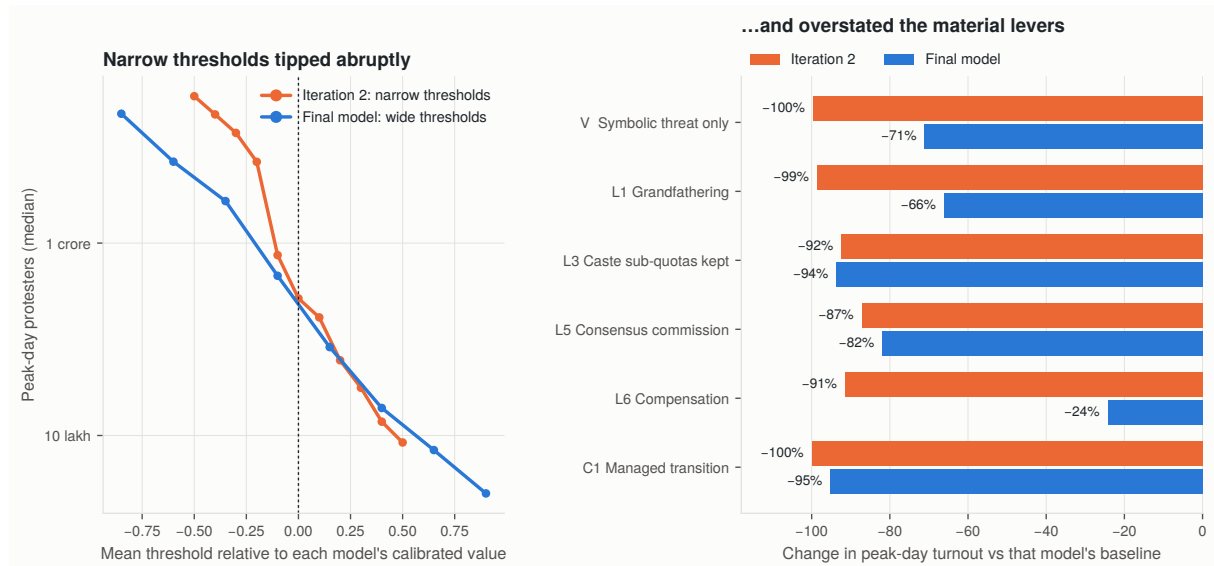


Figure 3: Left: peak turnout against mean threshold, centred on each model’s calibrated value. Right: iteration 2 overstated material levers and understated symbolic-only protest.

Lessons. Iteration 2 matched the turnout target but could not reproduce a documented episode, so matching a turnout target did not validate the model. Iteration 1 showed that each positive feedback in the model needs a bound. The spread of thresholds is a substantive assumption: with a narrow spread, a small change in grievance switches protest on or off. Some parameters are not separately identified: only the product λw_m enters grievance, and the ratio w_s/w_m is not identified by any turnout target (Section 11.7). The grounded specification was built on the same lessons. A single turnout target and a single validation episode are a weak test, so the grounded model is confronted with several episodes, including a null episode and one in which the upper castes, not SCs and STs, mobilized.

9 Experimental design

Table 7 maps each scenario to its parameter changes. The reference values are those of the stylized specification; Section 11.6 replaces each with a prior. In the grounded specification, material levers act on the segment table (grandfathering scales every loss by 0.3; compensation halves the losses of the three above-line losing segments), and L2 and L3 take their material changes from the allocation model (Section 5). In a package, allocation-rule levers are applied first, so that levers that scale losses act on the rule actually adopted.

Table 7: Scenarios and how they enter the model (reference values).

Code	Scenario	Parameter changes
Base	Abrupt ₹8L income-only switch	None
V	Symbolic threat only (validation)	All material changes set to 0
L1	Grandfather current cohorts, 10-year glide	All material losses $\times 0.3$; $s_g \times 0.95$
L2	Expand seats by a quarter	Stylized: below-line SC/ST/OBC losses set to 0. Grounded: allocation model with 25% more seats
L3	Keep caste quotas, income filter inside them	$s_g \times 0.4$ for SC, ST, OBC. Stylized: below-line losses 0. Grounded: allocation model with SC/ST quotas closed to above-line families
L4	Sub-classify to favour the most-deprived	Most-deprived SC/ST tier: material change reduced by 0.5 (half the unit loss); its symbolic threat falls from 0.8 to 0.3 times the group's
L5	Data-first commission, cross-party consensus	a : 1.5 to 1.0; $s_g \times 0.8$
L6	Compensate above-line losers	Above-line losses $\times 0.5$
L7	Guarantee untouched protections	$s_g \times 0.85$
B1	Internet shutdowns	$\beta_n, \beta_v \times 0.6$; bandh mobilization $\times 0.8$; violent deaths $\times 2.5$
B2	Heavy policing and mass arrests	Turnout cost 0.4 on bandh days; police-attributed deaths raised so total deaths $\times 3$; martyr response $\times 1.5$
C1	Managed transition	L2+L1+L5+L6+L7
C2	Hybrid package	L3+L4+L1+L5+L6+L7

The main comparison runs every scenario 50 times with paired draws in the grounded specification, and 50 times in each of two regimes of the stylized specification (central, and a high-mobilization regime with $\bar{\theta}$ lowered by 0.5). The outcomes are peak-day protesters, unique participants over the 40-day campaign, protester-days, and deaths. The high-mobilization regime is reported in Appendix E.

10 Calibration results

Retained parameter sets. Of 6,000 parameter sets drawn from the priors, 19 were not ruled out by the event counts, deaths and anchors. After 3 waves, 281 sets survive (Figure 4). By construction, every survivor reproduces every target within the stated uncertainty, including the September 2018 upper-caste bandh, which the model generates from the same mechanisms with the threat pointing the other way. The fit to the EWS null episode is the loosest: the survivors' median event rate lies above the observed one, and the fit relies on the wide observation interval that the low precision of the EWS filter produces.

Identified parameters. The mean threshold is 7.06 (90% interval 5.35–8.35), close to the stylized model's 6.6, which was tuned to an independent turnout target. The 2018 shock is 1.59

(0.50–2.38) times the stylized reform’s symbolic threat. Relative to 2018, the 2024 shock has median 0.57 (0.06–1.89), the September 2018 upper-caste shock 0.52 (0.07–1.57), and the EWS amendment 0.43 (0.07–1.16). The medians order the episodes as reporting does, but every interval is wide. The share of single-group neighbourhoods is 0.78 (0.54–0.98): the data do not support complete segregation, but do not pin mixing down either.

Turnout and the death rate. The implied turnout on 2 April 2018 is 63 lakh, with a 90% interval of 7–539 lakh. Event counts trade off against the observation model’s scale and exponent, and the two anchors narrow that trade-off only partly. The event data rule out the lowest reading of the press reports (“thousands”), but they cannot distinguish a few lakh from a few crore. Calibrating to event data constrains relative magnitudes far better than absolute turnout. The bandh-day death rate is 60 per crore protester-days (5–769), far above the stylized model’s ordinary-day rate of 4, because 11 to 14 people died on a single day in 2018.

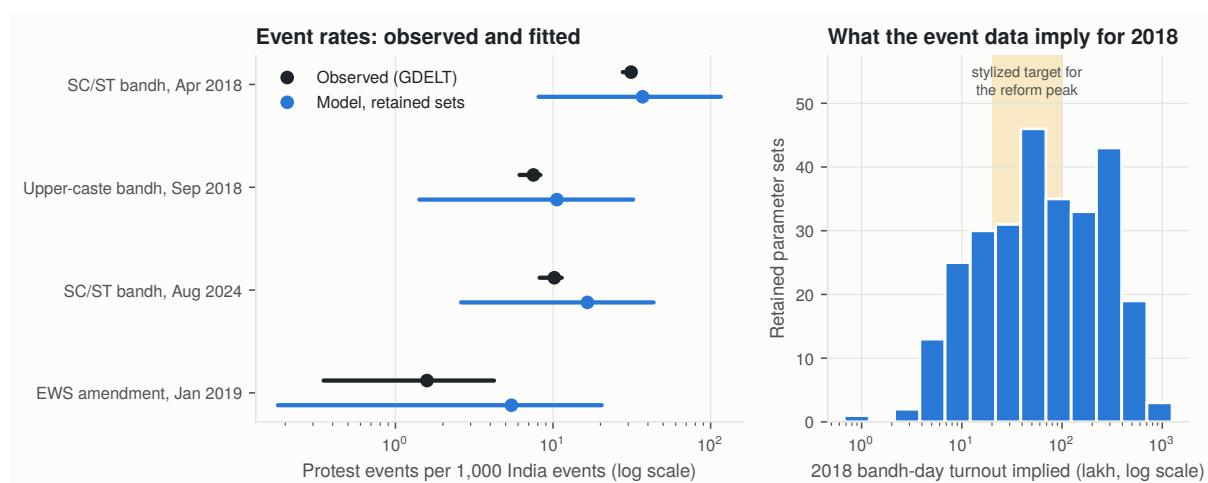


Figure 4: Left: observed event rates (median and 90% interval from the precision correction) and the rates produced by the retained parameter sets (median and 90% range). Right: 2018 bandh-day turnout implied by the retained sets, with the stylized model’s calibration range for the reform peak shaded for reference.

Out-of-sample tests. The leave-one-episode-out predictions were committed to the public repository, with their hashes, before they were scored (Table 8). Both the death and the spatial predictions are weak.

- **Deaths.** Without its own data, 2018’s bandh-day deaths are predicted at a median of 9 against 11–14 observed, but with an interval so wide as to be uninformative. For the two peaceful bandhs the model predicts deaths where none occurred. The model has one bandh-day death rate for all episodes, and the data say that violence is specific to the episode: it depends on policing, local counter-mobilization and chance in ways the model does not represent. The death outputs of the reform scenarios should be read in that light.
- **Geography.** Across 20 large states, the model’s distribution of protest correlates with the observed distribution of events at 0.57 for 2018 when that episode is held out, better than state population (0.38) and state SC-plus-ST population (0.49). For 2024 it is level with the SC-plus-ST baseline (0.73 against 0.72); for the upper-caste bandh it does worse than both baselines (0.42 against 0.60 and 0.63). Table 9 gives the same comparison with the full calibration. The district geography adds little predictive power beyond demography.

These results limit what the grounded specification can claim. It is grounded in the sense that its inputs come from data and its survivors reproduce four episodes’ event counts; it is not a

validated forecasting model of where, how large or how deadly protest will be.

Table 8: Leave-one-episode-out predictions, frozen and hashed before scoring. Deaths: observed range and predicted median with 90% interval of the mean. Spatial: Spearman correlation across large states between predicted protester-days and observed events, against two naive baselines.

Held-out episode	Sets	Deaths	Predicted	Inside	Model	Population	SC+ST
SC/ST Bharat Bandh, 2 April 2018	320	11–14	9.0 (0.0–1429.3)	yes	0.57	0.38	0.49
SC/ST Bharat Bandh, 21 August 2024	361	0	24.3 (0.1–1453.8)	no	0.73	0.57	0.72
Upper-caste bandh, 6 September 2018	373	0	18.8 (0.1–1678.8)	no	0.42	0.60	0.63

Table 9: Spatial test with the retained sets from the full calibration: Spearman correlation across large states between model protester-days and observed core-day events (median, 90% range over retained sets), against state population and state SC-plus-ST population.

Episode	Model	Population	SC+ST population
SC/ST Bharat Bandh, 2 April 2018	0.52 (0.43 to 0.60)	0.38	0.49
SC/ST Bharat Bandh, 21 August 2024	0.65 (0.51 to 0.74)	0.57	0.72
Upper-caste bandh, 6 September 2018	0.44 (0.26 to 0.62)	0.60	0.63

11 Results

11.1 The abrupt switch

In the grounded specification an abrupt switch produces a median peak of 2.14 crore (80% interval 30.6–1248.7 lakh), 3.32 crore participants and a median of 95 deaths. Organizations call the first bandh on day 3 in the median run, and the government concedes in 100% of runs, after a median of 5 days. The stylized specification, tuned to a peak of 20 lakh to 1 crore, gives 46.1 lakh.

The grounded level should not be read as a forecast. It is 1.59 times the stylized symbolic threat because that is what the 2018 episode implies, it inherits the wide range of 2018 turnouts that the event data allow, and it assumes that the reform shocks caste recognition as much as the 2018 ruling did ($R = 1$) and that opposition parties back the protest. Section 12.1 shows that the level moves by more than an order of magnitude across these assumptions. The comparisons between interventions, which are paired within each world, are the object of the analysis.

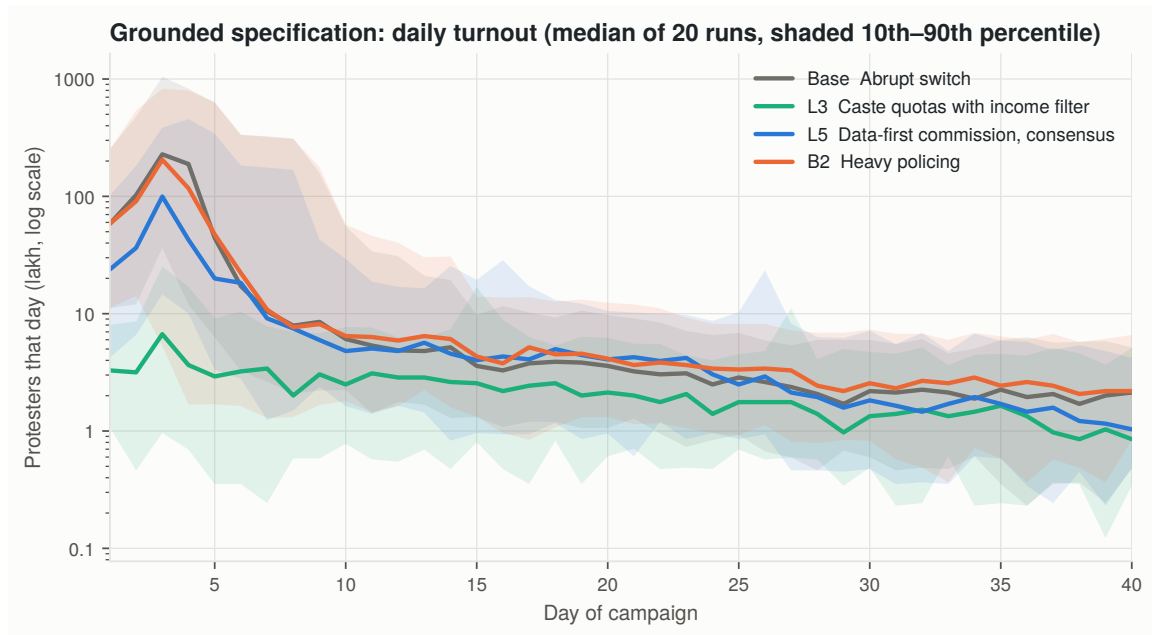


Figure 5: Grounded specification: daily turnout for four scenarios. Bandh days and concession are endogenous, so the median path is smoother than any single run.

Where the grievance comes from. Symbolic threat is 78% of all grievance that pushes agents towards protest (Figure 6). Material change is still large for the segments the allocation model identifies as losers: for below-line SC and ST agents it adds between a third and a half to their symbolic grievance. Removing all material change cuts the peak by -57% ; removing all symbolic threat cuts it by -99% . Symbolic threat drives more of the protest, and material loss still accounts for a large part of it.

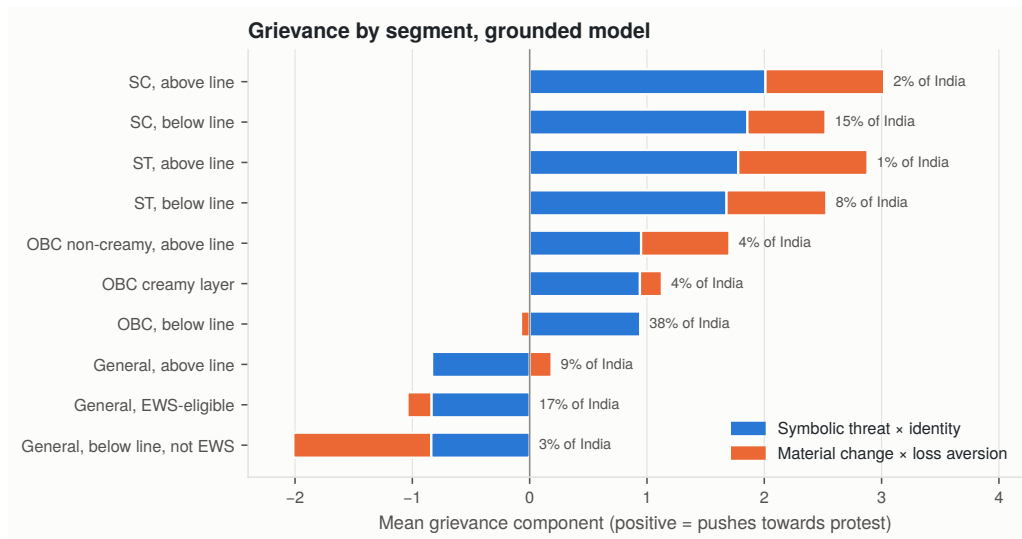


Figure 6: Mean grievance by eligibility segment in the grounded specification (central parameter values), split into symbolic threat and material change. Labels give each segment's share of the population.

Who protests and where. SC agents account for 63% of protester-days, ST agents for 26%, OBC agents for 7% and General agents, who counter-mobilize after a concession, for 2%. Among SC and ST agents, 9.5% of the better-off tier and 6.4% of the most-deprived tier protest at least once. Protest concentrates in the populous states with large SC populations: Uttar Pradesh

(15%), Bihar (8%), West Bengal (8%), Madhya Pradesh (7%), Maharashtra (7%), together 46% of protester-days (Figure 7). The distribution follows demography. It does not reproduce where the 2018 bandh was reported: Punjab, where the 2018 call began, had a far larger share of events than the model gives it, and West Bengal and Tamil Nadu far smaller ones. Local organization, which the model does not represent, shapes where a bandh takes hold; this is consistent with the weak spatial tests of Section 10.

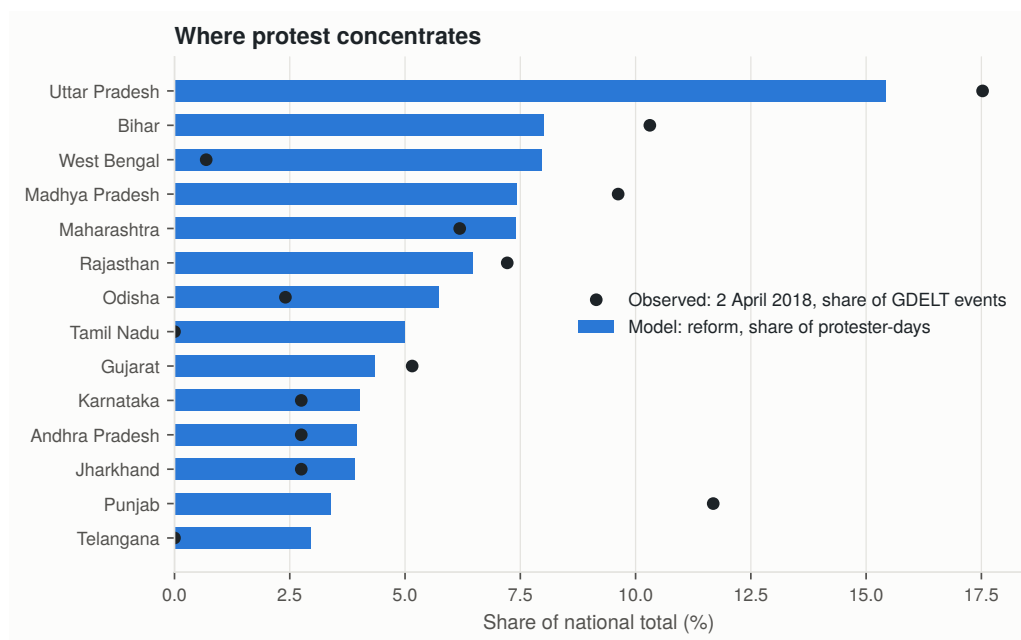


Figure 7: Share of protester-days by state under the abrupt switch (grounded specification, median over runs), and the share of GDELT protest events by state on 2 April 2018.

11.2 Stylized facts

Table 10 checks the grounded specification against the five stylized facts of Section 2.4. It reproduces S1, S3 and S4, and S2 largely. It reproduces S5 only in part: deaths occur and states concede, but the model cannot say which mobilizations turn violent.

Table 10: Stylized facts and the grounded specification.

Fact	Evidence from the model	Verdict
S1	Symbolic threat alone gives a peak of 64.1 lakh; the retained sets reproduce the 2018 event count from a symbolic shock with no material change.	Reproduced
S2	The peak falls on a bandh day in 66% of runs; turnout on bandh days is about 9 times that on ordinary days. In 18% of runs no bandh is held, mostly because the government concedes first, and in 16% turnout keeps building for a few days after the first bandh.	Largely reproduced
S3	Raising party backing from the 2018 level (0.2) to the 2024 level (0.6) changes the peak by a factor of 1.04; tripling the shock (from $R = 0.5$ to 1.5) by a factor of 18.	Reproduced: backing is a minor amplifier
S4	Sub-classification changes the share of the most-deprived tier that protests by -88% (6.4% to 0.8%) and of the better-off tier by -20% .	Reproduced
S5	Large simulated campaigns produce deaths (median 95 under the abrupt switch) and the government concedes in 100% of runs; but held-out predictions give deaths to peaceful bandhs (Table 8).	Partly: violence is not predicted

11.3 Interventions

Table 11 reports every scenario in the grounded specification, and Figure 8 compares its paired effects with the stylized specification's.

Table 11: Grounded specification, 50 paired runs. Peak and interval (10th–90th percentile) in lakh; change is the median paired change in peak turnout with a 95% bootstrap interval; ever protested in crore; deaths are medians.

	Scenario	Peak	10th–90th	Change	95% CI	Ever	Deaths
Base	Abrupt ₹8L income-only switch	214.1	30.6–1248.7	–	–	3.32	95
V	Validation: symbolic threat only (2018-type shock)	64.1	10.6–719.8	–57%	–63 to –49%	1.67	67.5
L1	Grandfather current cohorts + 10-year glide	83.5	13.6–751.2	–52%	–59 to –47%	1.69	70
L2	Expand seats by a quarter (EWS precedent)	193.4	27.1–1193.6	–9%	–12 to –8%	2.99	93.5
L3	Keep caste sub-quotas, add income filter	12.5	1.1–34.1	–97%	–98 to –94%	0.39	10
L4	Sub-classify to favour most-deprived	127.7	25.6–751.4	–40%	–41 to –38%	2.23	99.5
L5	Data-first commission + cross-party consensus	84.4	11.8–700.9	–58%	–61 to –53%	1.57	57.5
L6	Compensate above-line losers	197.0	28.2–1204.4	–8%	–9 to –6%	3.00	95
L7	Guarantee untouched protections	114.0	22.2–846.8	–40%	–44 to –35%	1.92	78.5
B1	Internet shutdowns	146.5	26.5–793.8	–26%	–28 to –23%	2.60	139
B2	Heavy policing and mass arrests	188.5	24.4–1167.5	–13%	–17 to 0%	3.41	219.5
C1	Managed transition package (L1+L2+L5+L6+L7)	18.7	1.9–78.6	–92%	–94 to –90%	0.70	15
C2	Hybrid design package (L3+L4+L1+L5+L6+L7)	1.7	0.4–12.5	–99%	–100 to –98%	0.13	2

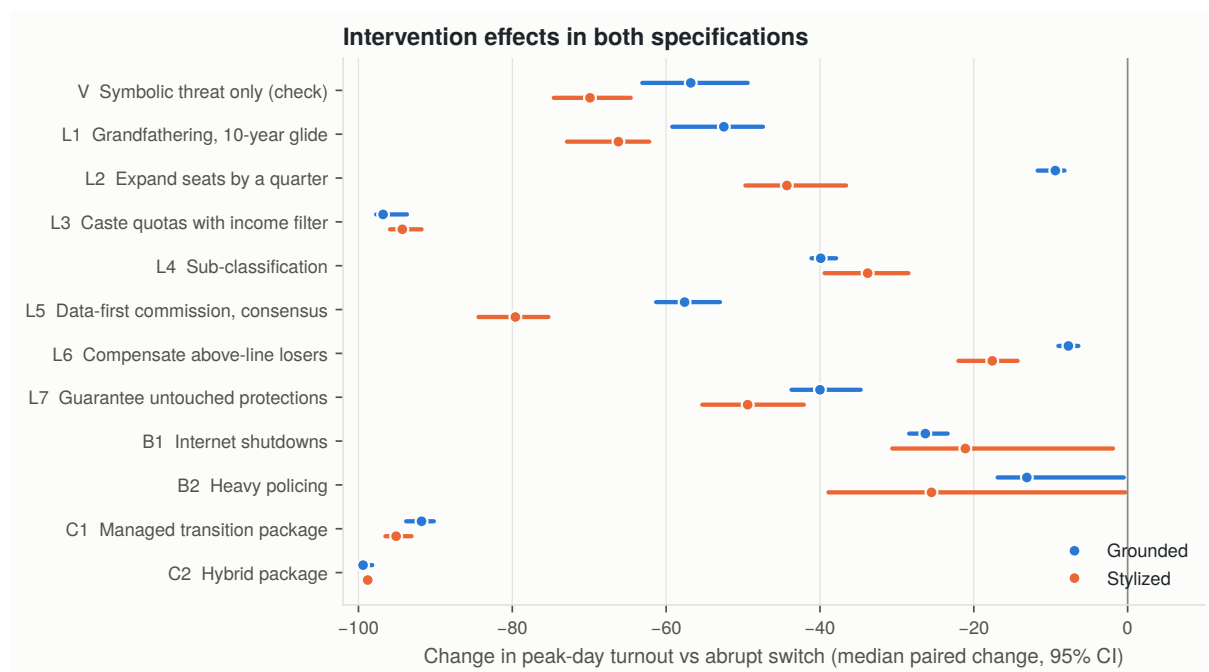


Figure 8: Median paired change in peak-day turnout with 95% bootstrap intervals, grounded and stylized (central regime) specifications.

Keeping caste as the qualifier dominates. Keeping caste quotas and adding an income filter inside them (L3) changes the grounded peak by -97% (95% CI -98 to -94%), and the stylized peak by -94% . In the grounded specification the lever lowers symbolic threat and, according to the allocation model, turns the reform's large seat loss for below-line SC and ST candidates into a gain (Section 5). It is also the lever furthest from the reform. It keeps caste as a criterion, so it is an alternative to the reform rather than a way of carrying it out.

Among levers that keep the income-only end state, process and phasing come first. A data-first commission with cross-party consensus (L5) changes the peak by -58% (-61 to -53%), and grandfathering with a ten-year glide (L1) by -52% (-59 to -47%). Their order is close in the grounded specification, and Section 11.6 shows that it depends on the priors over how strongly each lever acts. Statutory guarantees (L7, -40%) and sub-classification (L4, -40%) follow.

Seat expansion and compensation do little. Expanding seats by a quarter (L2) changes the grounded peak by only -9% , against -44% in the stylized specification. The stylized specification assumed that expansion removes the below-line loss; the allocation model shows that a merged pool with a quarter more seats still takes most of their seats from below-line SC and ST candidates (-63% for SC). The EWS precedent does not transfer, because EWS added a category without merging any. Compensating above-line losers (L6) is weakest (-8%), because it reaches a small minority and does not touch symbolic threat.

Packages. The managed transition (C1) keeps the income-only end state and changes the peak by -92% (-94 to -90%); the hybrid package (C2) by -99% .

Material grievance matters more than the stylized model assumed. Removing all material change (V) changes the grounded peak by -57% , against -70% in the stylized specification. The allocation model makes the material loss larger and broader than the stylized specification assumed, because below-line SC and ST candidates lose most of their seats, not a small share. Symbolic threat remains the larger driver, by a smaller margin.

11.4 Suppression

Internet shutdowns (B1) change the grounded peak by -26% and multiply deaths by 1.4; heavy policing (B2) changes it by -13% (-17 to 0%) and multiplies deaths by 2.6. Part of each death ratio is an input: B1 assumes 2.5 times as many violent deaths and B2 three times as many police-attributed deaths. Section 11.5 separates the two. Unlike the stylized specification, where heavy policing raises cumulative participation by $+14\%$, the grounded specification shows a change of -4% : campaigns end in concession within days, so the martyr effect has little time to act. The model therefore does not establish whether suppression broadens participation. Suppression raises deaths in every specification, but that follows from the assumed rise in deaths.

11.5 What is input and what is emergent

Every lever changes more than one input. Re-running each lever with one channel at a time, on the same paired worlds, separates them (Table 12).

- **L3** works through both of its channels. Its symbolic channel alone changes the peak by -90% and its allocation channel alone, which turns below-line seat losses into gains, by -53% .

- **L5's** effect comes mostly from its assumed 20% cut in symbolic threat (−48%); withdrawing party machinery alone gives −15%. Given S3, the symbolic channel is the more plausible of the two, but its size is a prior, not an estimate.
- **L1** works almost entirely through material loss (−46%; its 5% symbolic cut gives −13%).
- **B2.** The turnout cost alone changes the peak by −18%; the extra deaths and martyr response alone change it by +11% and deaths by +189%. The rise in deaths under heavy policing is almost all input: the scenario assumes three times as many police-attributed deaths. **B1** without its assumed extra violence changes the peak by −28% and deaths by −24%; with it, deaths change by +42%.

The stylized specification gives the same qualitative decomposition (Appendix G).

Table 12: Channel decomposition, grounded specification, 50 paired runs. Changes are median paired changes against the abrupt switch.

Lever	Channel	Peak	95% CI	Ever	Deaths
Abrupt switch	No material change	−57%	−63 to −49%	−46%	−50%
Abrupt switch	No symbolic threat	−99%	−99 to −98%	−96%	−98%
L1	Full lever	−52%	−59 to −47%	−42%	−46%
L1	Material channel only	−46%	−53 to −41%	−38%	−39%
L1	Symbolic channel only	−13%	−16 to −12%	−12%	−10%
L3	Full lever	−97%	−98 to −94%	−91%	−94%
L3	Symbolic channel only	−90%	−93 to −88%	−82%	−87%
L3	Allocation channel only	−53%	−58 to −46%	−42%	−44%
L5	Full lever	−58%	−61 to −53%	−45%	−52%
L5	Party amplifier only	−15%	−23 to −13%	−10%	−15%
L5	Symbolic channel only	−48%	−54 to −42%	−40%	−44%
B1	Full lever	−26%	−28 to −23%	−20%	+42%
B1	Coordination only (no extra violence)	−28%	−31 to −25%	−23%	−24%
B2	Full lever	−13%	−17 to 0%	−4%	+163%
B2	Turnout cost only	−18%	−25 to −13%	−8%	−18%
B2	Extra deaths and martyr response only	+11%	+6 to +17%	+9%	+189%

11.6 Uncertainty over how interventions act

The reference effect sizes in Table 7 are judgements. Each was given a uniform prior (for example, the commission's cut in symbolic threat from 0 to 30% and its cut in the party amplifier from 10 to 50%; the full list is in `protest_simulation/intervention_priors.py`). Each run draws one set of effect sizes and applies it to every lever in that run's world. Table 13 gives, for each specification, the paired effect and the probability that each lever ranks first to last; Figure 9 summarizes the probability of ranking first.

Table 13: Lever effects and rank probabilities under priors over effect sizes (200 runs, grounded specification; stylized in Appendix D). Change: median paired change in peak with 95% CI; rank 1 = largest reduction.

Lever	Change	95% CI	1	2	3	4	5	6	7
L1	−58%	−61 to −55%	3%	48%	30%	12%	6%	0%	1%
L2	−7%	−8 to −6%	0%	0%	1%	2%	8%	44%	44%
L3	−89%	−92 to −87%	95%	2%	1%	0%	0%	0%	1%
L4	−33%	−35 to −32%	0%	7%	15%	38%	36%	2%	1%
L5	−42%	−48 to −39%	1%	28%	34%	16%	16%	2%	2%
L6	−8%	−8 to −7%	0%	2%	1%	3%	10%	46%	38%
L7	−32%	−35 to −27%	0%	12%	18%	28%	24%	4%	12%

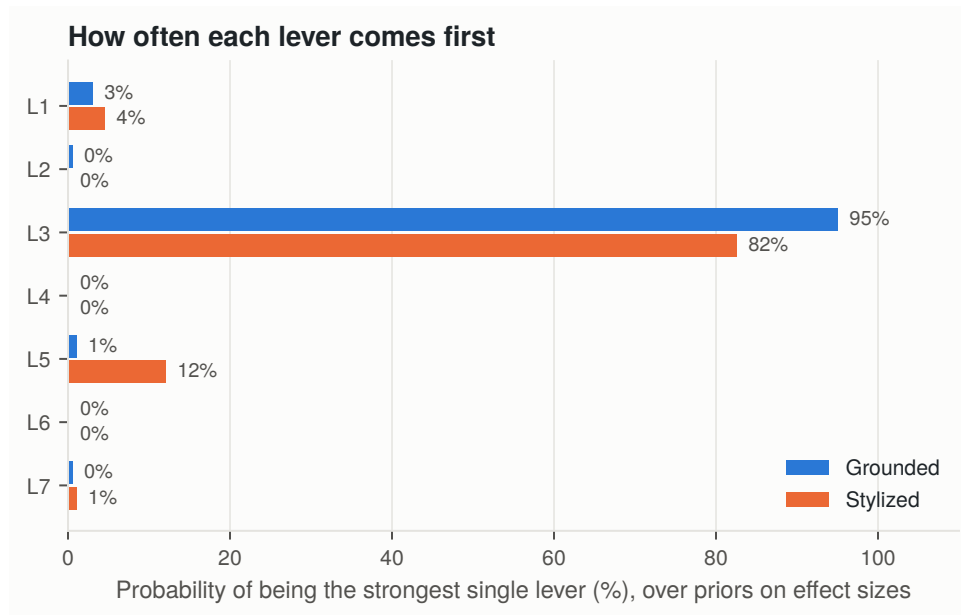


Figure 9: Probability that each single lever gives the largest reduction in peak turnout, over the priors on effect sizes.

L3 is the strongest single lever with probability 95% in the grounded specification and 82% in the stylized one. The commission beats grandfathering with probability 34% in the grounded specification and 58% in the stylized one. The reference values in Table 7 sit near the strong end of the commission's prior, which is why it ranks above grandfathering in Table 11; averaged over the priors, grandfathering is the stronger lever more often in the grounded specification. The order of the two is not settled by the model. Compensation is in the bottom two in 84% of grounded draws and 94% of stylized draws; in the grounded specification seat expansion shares the bottom (88%).

For suppression, the prior on the response of turnout to deaths runs from deterrence to strong backfire. Heavy policing raises cumulative participation in 22% of grounded runs and 37% of stylized runs; in the stylized specification the share rises from 0% in the quarter of runs with the weakest response to 74% in the quarter with the strongest. Backfire on participation is therefore a consequence of the assumed response to deaths, which no episode identifies.

11.7 Material versus symbolic weight

The ratio of the symbolic to the material grievance weight is not identified by any turnout target, and the episodes involve no material change, so they do not identify w_m either. In the stylized specification, a sweep that holds $w_s w_m$ fixed and recalibrates $\bar{\theta}$ at each point finds that a material lever is strongest only when w_s/w_m is at most 1.47, against the reference 7.3; the 2018-type validation check admits ratios 3.67–22.00. In the grounded specification, where $\bar{\theta}$ comes from the episodes, we multiply w_m directly (Table 14). No material lever becomes the strongest at any multiplier up to 8. Grandfathering overtakes the commission once w_m is doubled, and compensation stops being the weakest lever once w_m is quadrupled; grandfathering approaches L3 only at the largest multiplier, where the baseline peak is implausibly large. A heavier material weight reorders the middle of the ranking and leaves L3 first.

Table 14: Grounded specification: paired change in peak by lever as the material weight is multiplied ($R = 1$, backing 0.6).

w_m multiplier	Baseline peak	L1	L2	L3	L5	L6	L7	First
0.25	88.3 lakh	−26%	−2%	−94%	−63%	−2%	−44%	L3
0.5	1.23 crore	−35%	−5%	−95%	−62%	−4%	−42%	L3
1	2.14 crore	−52%	−9%	−97%	−58%	−8%	−40%	L3
2	3.94 crore	−71%	−13%	−96%	−43%	−13%	−29%	L3
4	15.49 crore	−79%	−13%	−92%	−25%	−14%	−14%	L3
8	36.14 crore	−86%	−10%	−89%	−6%	−14%	−3%	L3

11.8 Verdicts on the hypotheses

Table 15 summarizes the evidence on H1–H9 from the grounded specification, the priors over effect sizes and the robustness analyses.

Table 15: Hypotheses and verdicts. Change: median paired change in peak turnout, grounded specification. Rank: probability of being the strongest single lever, and of being in the bottom two, under the priors over effect sizes.

	Hypothesis	Change	Rank (first / bottom two)	Verdict
H1	Grandfathering reduces protest	−52%	3% / 2%	Supported; acts through material loss
H2	Expanding seats reduces protest	−9%	0% / 88%	Weak: a merged pool still takes most below-line SC/ST seats
H3	Caste quotas with an income filter reduce protest	−97%	95% / 1%	Strongly supported, through both symbolic and allocation channels
H4	Sub-classification splits the coalition	−40%	0% / 4%	Supported: most-deprived participation changes by −88%
H5	A consensus commission reduces protest	−58%	1% / 4%	Supported, mostly through its assumed symbolic effect; order against H1 unsettled
H6	Compensating direct losers reduces protest	−8%	0% / 84%	Weak
H7	Statutory guarantees reduce protest	−40%	0% / 17%	Supported
H8	Internet shutdowns reduce protest	−26%		– Lower peak, more deaths ($\times 1.4$); not a mitigation
H9	Heavy policing reduces protest	−13%		– Small, uncertain reduction; deaths $\times 2.6$; backfire depends on the response to deaths

12 Robustness

12.1 The grounded specification's unidentified assumptions

Table 16 varies the reform's shock relative to 2018 ($R = 0.5, 1, 1.5$) and the party backing of the protest (0.2, the 2018 level, and 0.6, the 2024 level). The baseline peak ranges from 29.0 lakh to 6.78 crore. L3 is the strongest single lever in 6 of 6 cells, and compensation is the weakest in 6.

Table 16: Grounded specification under alternative values of the reform's shock (R) and party backing; 50 paired runs per cell.

R	Backing	Peak	Deaths	Conceded	L1	L3	L5	L6	First
0.5	0.2	29.0 lakh	25	88%	−66%	−88%	−53%	−11%	L3
0.5	0.6	37.0 lakh	28	88%	−64%	−90%	−57%	−11%	L3
1.0	0.2	2.06 crore	92.5	98%	−53%	−97%	−56%	−8%	L3
1.0	0.6	2.14 crore	95	100%	−52%	−97%	−58%	−8%	L3
1.5	0.2	6.09 crore	182.5	100%	−43%	−97%	−48%	−5%	L3
1.5	0.6	6.78 crore	209	100%	−42%	−97%	−49%	−5%	L3

12.2 Structural ensemble

Each variant in Table 17 switches on one structural alternative, or all of them, recalibrates $\bar{\theta}$ so that the stylized baseline again hits its turnout target, and reruns every lever with 100 paired runs. L3 is strongest in 11 of 11 variants and compensation weakest in 11; L5 beats L1 in 11. Heavy policing raises cumulative participation in 10 variants, and not when deaths deter.

Table 17: Structural ensemble (stylized specification, recalibrated): paired change in peak for four levers, change in cumulative participation under heavy policing, and the strongest / weakest single lever.

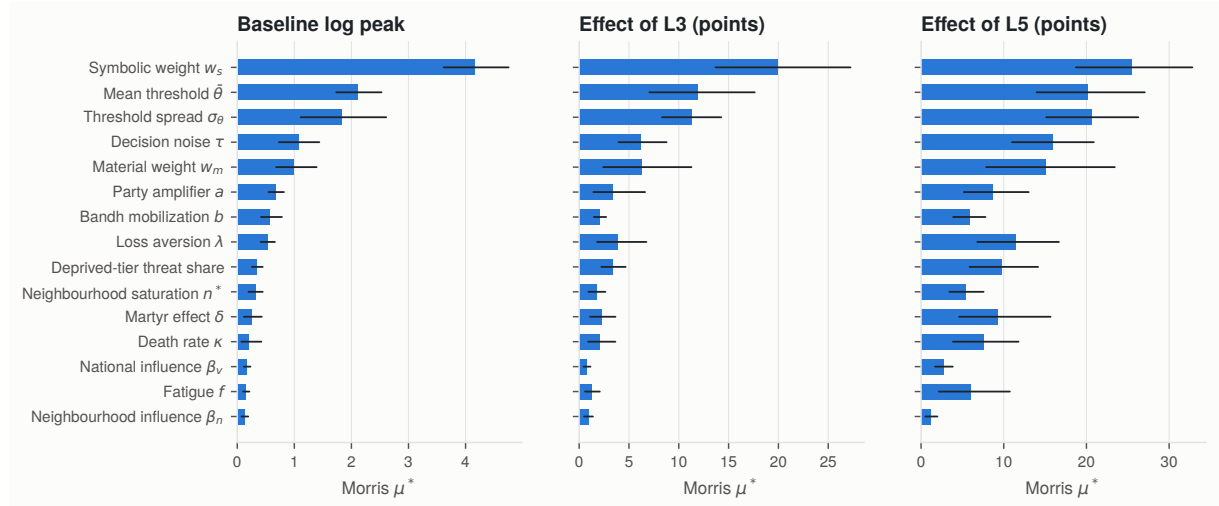
Variant	$\bar{\theta}$	L1	L3	L5	L6	B2 cum.	First / last
Reference structure	6.60	−69%	−95%	−81%	−18%	+14%	L3 / L6
Separate random streams for decisions, deaths and events	6.64	−72%	−94%	−82%	−21%	+17%	L3 / L6
Logistic (heavier-tailed) threshold distribution	6.93	−49%	−82%	−63%	−12%	+7%	L3 / L6
3% activist core with low thresholds	7.05	−60%	−94%	−74%	−12%	+15%	L3 / L6
Organizations call bandhs when turnout is credible	6.67	−81%	−98%	−86%	−20%	+10%	L3 / L6
Government may concede; General category counter-mobilizes	6.54	−62%	−93%	−73%	−18%	+5%	L3 / L6
Negative-binomial deaths, split into police-attributed and other	6.63	−68%	−94%	−78%	−19%	+18%	L3 / L6
Deaths deter rather than mobilize	6.24	−54%	−88%	−68%	−15%	−40%	L3 / L6
30% of neighbourhoods mixed across groups	6.59	−68%	−95%	−80%	−18%	+14%	L3 / L6
Half of above-line OBC non-creamy today; 15% of below-line General fail EWS asset tests	6.61	−69%	−94%	−81%	−19%	+18%	L3 / L6
All alternatives above except deterrence	6.95	−54%	−97%	−65%	−15%	+9%	L3 / L6

12.3 Global sensitivity

Morris screening over fifteen behavioural parameters (20 trajectories, 320 design points, stylized specification) finds that the symbolic weight dominates the baseline peak and every lever's effect, followed by the mean threshold and the threshold spread (Table 18, Figure 10). Across all design points, L3 cuts the peak more than L5 in 98% and L5 more than L1 in 75%. The same screening around the grounded specification's central calibrated values (Appendix H) ranks Symbolic weight w_s first for the baseline peak and Threshold spread σ_θ first for L3's effect; L3 beats L5 in 100% of its design points and L5 beats L1 in 61%.

Table 18: Morris screening: mean absolute elementary effect μ^* on the baseline log peak and on the paired peak change (percentage points) of L1, L3 and L5, and σ for the log peak.

Parameter	Log peak	L1	L3	L5	σ (log peak)
Symbolic weight w_s	4.16	26	20	25	1.38
Mean threshold $\bar{\theta}$	2.11	24	12	20	0.94
Threshold spread σ_θ	1.83	26	11	21	1.73
Decision noise τ	1.08	17	6	16	0.81
Material weight w_m	0.99	28	6	15	0.82
Party amplifier a	0.68	9	3	9	0.32
Bandh mobilization b	0.57	8	2	6	0.44
Loss aversion λ	0.53	15	4	11	0.29
Deprived-tier threat share	0.34	10	3	10	0.23
Neighbourhood saturation n^*	0.32	5	2	5	0.31
Martyr effect δ	0.26	9	2	9	0.43
Death rate κ	0.20	8	2	8	0.46
National influence β_v	0.17	4	1	3	0.14
Fatigue f	0.16	3	1	6	0.13
Neighbourhood influence β_n	0.13	2	1	1	0.13

**Figure 10:** Morris μ^* with 95% bootstrap intervals over trajectories.

12.4 One-at-a-time sensitivity, L3's symbolic assumption, and convergence

Each of twelve parameters perturbed by $\pm 25\%$ (or ± 0.25 for $\bar{\theta}$) changes the ranking of levers little: the Spearman correlation with the reference ranking has a median of 1.00 and a minimum of 0.93, and L3 is strongest in 24 of 24 perturbations (Table 19). L3 rests on the assumption that retaining caste categories removes 60% of symbolic threat; the 2024 bandh, triggered by a suggested SC/ST creamy layer, is reason for caution. Leaving 40%, 60% and 80% of the threat in place gives stylized peaks of 2.7 lakh, 4.6 lakh and 9.2 lakh. Medians show no systematic trend with agent count between 30,000 and 240,000 agents (Table 20).

Table 19: One-at-a-time sensitivity (stylized). Peaks in lakh (20 runs); ρ for lowered and raised parameter; strongest and weakest single lever (lowered/raised).

Parameter	Lowered	Raised	ρ_-	ρ_+	Strongest	Weakest
Loss aversion	32.4	85.1	0.98	0.98	L3/L3	L6/L4
Material-loss weight	32.4	85.1	0.98	0.98	L3/L3	L6/L4
Symbolic-threat weight	10.6	230.1	0.98	0.98	L3/L3	L6/L6
Opposition-party amplifier	26.8	78.1	1.00	1.00	L3/L3	L6/L6
Neighbourhood influence	43.5	46.2	1.00	1.00	L3/L3	L6/L6
National visibility influence	42.5	46.5	1.00	1.00	L3/L3	L6/L6
Threshold spread	5.8	222.3	0.98	0.93	L3/L3	L4/L6
Decision noise	30.0	84.9	1.00	1.00	L3/L3	L6/L6
Fatigue	56.4	40.6	1.00	1.00	L3/L3	L6/L6
Death rate	37.5	69.8	1.00	1.00	L3/L3	L6/L6
Martyr effect per death	48.0	46.7	1.00	1.00	L3/L3	L6/L6
Mean threshold (± 0.25)	117.0	18.9	1.00	1.00	L3/L3	L6/L6

Table 20: Agent-count convergence (stylized; 20 paired runs each).

Agents	People per agent	Baseline peak (lakh)	10th–90th	L3 peak (lakh)
30,000	48,700	53.0	11.1–119.6	2.4
60,000	24,300	50.1	10.7–122.2	1.9
120,000	12,200	42.7	14.2–109.4	2.6
240,000	6,100	53.7	11.2–120.0	2.3

13 Discussion

Conclusions that changed. Comparing the grounded specification with the stylized one shows which conclusions depended on assumptions. The stylized specification treated the reform as a small eligibility change confined to SC and ST families above the line. Under the actual creamy-layer and EWS rules, 6.8% of Indians lose eligibility, and in IIT admissions below-line SC and ST candidates, who keep eligibility, lose most of their seats to below-line General candidates. The stylized specification also assumed that seat expansion on the EWS model removes the below-line loss; a merged pool still ranks SC and ST candidates against General candidates, so expansion does little. Its finding that heavy policing raises total participation depends on an assumed response of turnout to deaths, and does not hold in the grounded specification, where campaigns end in concession quickly. Material grievance also matters more than the stylized specification assumed, because the seat losses of below-line SC and ST candidates are large.

Conclusions that held. The ordering at the top and bottom of the lever ranking survived every analysis in this paper: keeping caste as the qualifier with an income filter inside it (L3) cuts protest most, and compensating the direct losers (L6) is among the least effective, joined in the grounded specification by seat expansion (L2). L3 is first in both specifications, across the priors over effect sizes, across the structural variants, across the Morris design, and across the grounded specification's unidentified shock size, party backing and material weight. Compensation is last or second to last in almost all of them; the exception is a material weight at least four times its reference value, which implies turnouts far beyond any recorded protest. Most of the mobilizable grievance is symbolic and spread across the SC and ST population, and in the grounded specification the material loss is also concentrated on below-line SC and ST candidates; L3 removes much of both, while compensation reaches a small minority and touches

neither.

Open question: commission or grandfathering. Among levers that keep the income-only end state, the order of the data-first commission (L5) and grandfathering (L1) is not settled. At the reference values the commission is ahead; averaged over the priors on how strongly each lever acts, grandfathering is ahead more often (the commission wins with probability 34%). The order depends on how strongly a commission would lower symbolic threat, on the size of the reform's shock (grandfathering comes first when $R = 0.5$), and on the material weight (grandfathering comes first once w_m is doubled), and none of these is identified. The decomposition shows that most of L5's effect in the model comes from its assumed cut in symbolic threat. Given that the 2018 bandh was large without party backing, the claim that consensus works by withholding party machinery is the weaker half of the mechanism. A commission's value, if it has one, would lie in changing what the reform means to those it affects.

Compensation and the retrenchment literature. Pierson (1994) treats compensation as a blame-avoidance tool. Its value depends on how grievance is composed. Where losses are material and the losers are the mobilizers, compensation addresses the grievance. Where the grievance is largely symbolic, or where the material loss falls on a broad group that compensation schemes do not reach, it does not.

Policy implications. Conditional on the model:

1. Whether caste stays a criterion matters more than any compensation scheme. An income filter inside caste categories achieves most of the income targeting that motivates the reform, at a small fraction of the unrest, and in IIT admissions it moves seats towards below-line SC and ST candidates rather than away from them.
2. A merged income-only pool is not a neutral re-labelling. It moves most reserved seats of below-line SC and ST candidates to below-line General candidates, and expanding capacity on the EWS model does not undo that.
3. If the income-only end state is the goal, phasing (grandfathering) and a process that changes what the reform signals (a commission, statutory guarantees) matter most, and they compound.
4. Suppression trades a lower peak for more deaths. Whether it also broadens protest depends on how deaths are received, which the model cannot tell.

Where protest would happen. The model places the reform's protest where SC and ST populations are largest, but the 2018 record shows that local organization can shift where a bandh concentrates: Punjab, where the 2018 call began, reported far more events than its population share. The model's state distribution is a demographic baseline, not a prediction of where organizers will succeed.

Feasibility. The levers that work best are also the hardest to obtain or the furthest from the reform as proposed. Cross-party consensus on reservation is rare; L3 keeps caste as a criterion. The model treats political feasibility as exogenous.

What the platform can be used for. The same code can answer questions beyond this paper's. The allocation model evaluates any pool design under the over-and-above rule, for example different pool sizes, caste sub-quotas within an income pool, or a creamy layer for SC and ST; the scenario catalogue takes new interventions as short functions; and the calibration machinery can absorb new episodes or new event data as they become available.

14 Limitations and threats to validity

- **The reform's shock size is not identified.** The episodes identify how strongly the 2018 ruling threatened SC and ST recognition relative to the other episodes, not how strongly an income-only reform would. The grounded analysis assumes $R = 1$ and backs the reform with opposition parties at the 2024 level. Table 16 shows that the level of protest moves by an order of magnitude across these assumptions; the ranking moves much less.
- **The material weight is not identified.** The episodes involve no material change, so w_m is assumed in both specifications. The break-even analyses bound how far it would have to move for the ranking to change.
- **Event counts are an indirect measure.** GDELT counts reports, not people. The observation model is concave and anchored on two episodes with published crowd estimates, but the implied turnout for 2018 remains wide (7–539 lakh). The relevance labels were assigned from URL text by a single coder and have not been independently double-coded; the precision correction inherits that weakness. ACLED would provide an independent cross-check.
- **Party backing and shock direction are coded, not estimated.** Each episode's party backing and the direction of its symbolic threat are judgements from reporting, held fixed in calibration; errors are absorbed by the estimated magnitudes.
- **Intervention mappings are priors.** How much a commission, a guarantee or a filter reduces threat is not observed. The rank probabilities integrate over wide priors, but they cannot correct a mapping whose whole range is wrong.
- **The allocation model covers one institution type.** Seat changes are estimated for IIT admissions, where rank lists are public. Government jobs, state institutions and medical admissions may differ; NEET failed our tail check. Family income is not observed per candidate and enters through priors constrained by aggregate evidence. Horizontal reservations and programme preferences are not modelled.
- **Omitted structure.** Jati-level heterogeneity beyond two SC/ST tiers, state-level variation in income and organization, media dynamics, courts and electoral feedback are not modelled. State government alignment was coded (data/derived/state_government_alignment.csv) but is not used by the model.
- **Generic dynamics.** Threshold cascades can produce heavy-tailed outcomes whatever their micro-foundations (Thron and Jackson, 2015). The episode tests, the structural ensemble and the separation of inputs from emergent effects are partial answers; they do not establish that the mechanisms are the right ones.

15 Conclusion

Replacing caste-based reservation with a single income test at ₹8 lakh changes more than the eligibility of SC and ST families above the line. Under the actual rules, 6.8% of Indians would lose eligibility and 3.1% would gain it, and in a merged pool it moves most seats held by below-line SC and ST candidates to below-line General candidates, although those candidates keep their eligibility.

How much protest that would provoke is not something the available data can pin down. Event counts from past episodes identify the relative size of past shocks, not their headcounts, and out-of-sample predictions of deaths and geography are weak. The model can compare designs on the same simulated worlds, and the main lines of that comparison hold across every analysis. Keeping caste as the qualifier with an income filter inside it cuts protest most, in every

analysis in this paper, because it removes both the symbolic threat and, as the allocation model shows, the seat losses of the below-line majority. Compensating the direct losers and expanding seats on the EWS model do little. If the income-only end state is the goal, phasing and a process that changes what the reform means matter most, though the model cannot say which of the two matters more. Suppression costs lives in every specification, and its effect on the size of protest depends on how people respond to deaths, which no past episode identifies.

Future work. An independent event dataset (ACLED) would test the GDELT counts, the spatial predictions and, through its reported crowd sizes, the implied turnout of past bandhs; the pipeline for it is in place. A second, independent coding of the relevance audit would tighten the precision correction. Seat changes for government jobs and state institutions would test whether the IIT results generalize. Survey experiments on how an income filter, a commission or statutory guarantees change perceived threat would replace the priors over intervention effects with estimates.

16 Ethics statement

The study uses no individual-level data. The population is synthetic and built from published aggregate shares; the JEE rank lists are used without roll numbers or names, and only rank and category are retained. Suppression tactics are included only to test whether they backfire, and they are not presented as options. Obfuscation, which is manipulating what citizens know about a reform, is deliberately excluded from the intervention set. The model is not a forecast of any actual policy. It is intended to inform the design of reforms that respect democratic contestation, not to help any actor suppress it. The state-level results describe where protest would concentrate; they are not, and should not be used as, a basis for pre-emptive policing.

17 Reproducibility

All code, derived data, results, figures and videos are available at github.com/utsavbalhara/reservation-reform-protest-model. Seeds are fixed, and runs distributed across processes give results identical to serial runs. Every number in this paper is generated from the result files into LaTeX macros; none is typed by hand. A test suite checks the reference model against stored outputs and against an independent scalar reimplementations, and checks model properties (turnout moves in the expected direction with grievance and threshold, campaign accounting identities hold, each intervention touches only its own parameters, parallel and serial runs agree), and checks the choice rules, the rank-list extraction and the district population against their sources. The main commands, in order, are:

```
python -m data_pipelines.jee_advanced_rank_lists      # rank lists -> data/derived
python -m data_pipelines.gdelt_download               # GDELT daily files (raw, not stored)
python -m data_pipelines.gdelt_episode_events
python -m data_pipelines.episode_targets
python -m experiments.eligibility_accounting
python -m experiments.merged_pool_allocation --draws 300
python -m experiments.episode_calibration --wave-size 6000 --waves 3
python -m experiments.score_frozen_predictions
python -m experiments.compare_interventions --specification grounded
python -m experiments.compare_interventions --regime central
python -m experiments.compare_interventions --regime high-mobilization
python -m experiments.record_daily_trajectories --specification grounded
python -m experiments.grounded_channels
python -m experiments.grounded_assumptions
python -m experiments.intervention_mapping_uncertainty --specification grounded
python -m experiments.intervention_mapping_uncertainty --specification stylized
```

```
python -m experiments.material_symbolic_break_even --specification stylized
python -m experiments.decompose_levers
python -m experiments.global_sensitivity --specification grounded
python -m experiments.global_sensitivity --specification stylized
python -m experiments.structural_ensemble
python -m experiments.stylized_facts
python -m experiments.sensitivity_analysis
python -m experiments.check_robustness
python -m experiments.replay_model_development
python -m visualization.render_grounded_figures
python -m visualization.write_latex_result_macros
python -m visualization.write_paper_macros
python -m visualization.write_grounded_macros
python -m pytest
```

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A Novelty search

Before claiming novelty we searched Google Scholar (18 queries combining “reservation”, “caste”, “quota”, “affirmative action”, “Bharat Bandh”, “Mandal” and “Dalit” with “agent-based”, “simulation”, “threshold model”, “computational model” and “protest”; first page of results each) and SSRN (through a web search restricted to ssrn.com, since SSRN’s own search refused automated access). The nearest hits were agent-based models of food riots, of ethnic minority rule and civil war, and of affirmative action in hiring and college sorting, and a model of reservation policy in education markets; none modelled protest against reservation or compared reform designs by the mobilization they would provoke. OpenAlex and Semantic Scholar were rate-limited and not used. These were keyword searches, not a systematic review. The queries, hit counts and titles are in docs/literature_search.md and docs/literature_search_scholar_results.json.

B Data sources and processing

- **JEE (Advanced) rank lists.** The Joint Implementation Committee reports (jeeadv.ac.in/reports/<year>.pdf) print the Common Rank List and each category’s rank list, sorted by roll number. Joining them on roll number gives the category of every candidate on the Common Rank List. Only rank and category are kept. The 2019 and 2020 lists fail a check that category ranks increase with Common Rank List rank and are not used. Registered, appeared and allotted counts by category are transcribed from the same reports; the 2024 appeared counts are imputed.
- **Census 2011.** District population and SC and ST counts from the Primary Census Abstract (640 districts; the ten Telangana districts are separated from Andhra Pradesh).
- **NFHS-5 (2019–21).** Caste of household head by state (India Report, Table 2.10), used for the OBC share of the non-SC, non-ST population.
- **GDELT 1.0.** Daily event files for each episode’s window; CAMEO root code 14, action located in India. Keyword filters and the exclusion keywords for each episode are in `data_pipelines/episodes.py`. The raw files are not stored in the repository; the pipeline downloads them.
- **Relevance audit.** 246 matched source URLs, sampled from core and non-core days of every episode, were labelled as relevant or not from the article’s URL text (headline slug), without opening the articles. Labels from URL text are a weaker basis than reading each article, and a second independent coding with an agreement statistic has not yet been done. The labels are in `data/derived/gdelt_relevance_audit.csv`.
- **Deaths and party backing.** Coded per episode from the cited reporting in `data_pipelines/episodes.py`.

C Allocation model: robustness

Table 21: Median seat change by segment across processing orders and pool sizes, and for the two allocation-rule levers (reserved-first order).

Order	Pool	SC above	SC below	ST below	OBC below	GEN-EWS	GEN below
reserved-first	59.5%	−96%	−72%	−85%	+23%	+71%	+158%
reserved-first	49.5%	−95%	−76%	−87%	+8%	+43%	+157%
reserved-first	69.5%	−98%	−68%	−82%	+39%	+101%	+156%
open-first	59.5%	−94%	−68%	−82%	+12%	+51%	+224%
open-first	49.5%	−92%	−71%	−83%	+4%	+41%	+208%
open-first	69.5%	−96%	−65%	−80%	+20%	+62%	+239%
L2: pool with 25% more seats		−95%	−63%	−79%	+57%	+116%	+211%
L3: caste quotas, income filter		−97%	+32%	+37%	0%	0%	−1%

D Stylized specification: rank probabilities

Table 22: Lever effects and rank probabilities under priors over effect sizes, stylized specification (200 runs).

Lever	Change	95% CI	1	2	3	4	5	6	7
L1	−69%	−72 to −67%	4%	35%	42%	18%	0%	0%	0%
L2	−37%	−40 to −35%	0%	0%	3%	30%	50%	18%	0%
L3	−88%	−90 to −87%	82%	14%	3%	0%	0%	0%	0%
L4	−31%	−34 to −28%	0%	0%	2%	13%	32%	43%	10%
L5	−73%	−76 to −69%	12%	44%	27%	10%	4%	4%	0%
L6	−18%	−21 to −16%	0%	0%	0%	0%	6%	28%	66%
L7	−48%	−55 to −42%	1%	8%	24%	29%	8%	10%	21%

E Stylized specification: central and high-mobilization regimes**Table 23:** Stylized specification, central regime, 50 paired runs. Columns as in Table 11.

	Scenario	Peak	10th–90th	Change	95% CI	Ever	Deaths
Base V	Abrupt ₹8L income-only switch	46.1	14.9–118.9	–	–	1.29	18.5
	Validation: symbolic threat only (2018-type shock)	13.3	6.0–37.6	–70%	–75 to –65%	0.47	5.5
L1	Grandfather current cohorts + 10-year glide	15.6	6.6–41.5	–66%	–73 to –62%	0.50	5.5
L2	Expand seats by a quarter (EWS precedent)	24.1	9.8–65.8	–44%	–50 to –37%	0.74	11
L3	Keep caste sub-quotas, add income filter	2.9	1.7–4.3	–94%	–96 to –92%	0.10	1
L4	Sub-classify to favour most-deprived	26.9	11.6–71.4	–34%	–39 to –28%	0.83	13.5
L5	Data-first commission + cross-party consensus	8.3	4.1–24.2	–80%	–84 to –75%	0.41	4.5
L6	Compensate above-line losers	34.9	13.1–93.9	–18%	–22 to –14%	1.08	16.5
L7	Guarantee untouched protections	18.2	9.6–55.2	–49%	–55 to –42%	0.68	7.5
B1	Internet shutdowns	42.8	17.2–73.7	–21%	–31 to –2%	1.41	54.5
B2	Heavy policing and mass arrests	32.9	17.4–73.3	–26%	–39 to 0%	1.61	79.5
C1	Managed transition package (L1+L2+L5+L6+L7)	2.2	1.4–3.4	–95%	–96 to –93%	0.12	1
C2	Hybrid design package (L3+L4+L1+L5+L6+L7)	0.6	0.4–0.9	–99%	–99 to –98%	0.04	0

Table 24: Stylized specification, high-mobilization regime ($\bar{\theta}$ lowered by 0.5), 50 paired runs. Peak and interval in lakh; change is the median paired change with 95% bootstrap interval; cumulative in crore.

	Scenario	Peak	10th–90th	Change	95% CI	Ever	Deaths
Base	Abrupt ₹8L income-only switch	209.4	92.6–388.4	–	–	4.14	79
V	Validation: symbolic threat only (2018-type shock)	66.1	18.0–152.5	–65%	–73 to –62%	1.84	27
L1	Grandfather current cohorts + 10-year glide	74.2	20.0–160.7	–62%	–68 to –59%	1.98	30.5
L2	Expand seats by a quarter (EWS precedent)	117.0	35.5–258.0	–46%	–49 to –41%	2.50	45
L3	Keep caste sub-quotas, add income filter	7.7	5.1–12.5	–96%	–97 to –96%	0.29	3
L4	Sub-classify to favour most-deprived	106.5	40.0–225.0	–44%	–48 to –40%	2.54	46.5
L5	Data-first commission + cross-party consensus	42.2	13.5–106.7	–78%	–80 to –74%	1.55	26.5
L6	Compensate above-line losers	167.0	49.3–327.2	–15%	–19 to –13%	3.80	65
L7	Guarantee untouched protections	110.8	34.9–247.2	–47%	–52 to –42%	2.52	43
B1	Internet shutdowns	110.6	55.9–184.4	–47%	–50 to –42%	3.43	152.5
B2	Heavy policing and mass arrests	124.3	52.4–245.6	–37%	–41 to –34%	4.20	254.5
C1	Managed transition package (L1+L2+L5+L6+L7)	6.4	3.6–10.0	–97%	–97 to –96%	0.36	4
C2	Hybrid design package (L3+L4+L1+L5+L6+L7)	1.9	1.3–2.9	–99%	–99 to –99%	0.12	1

F Model summary in the ODD format

Element	Content
Purpose	Compare designs of an income-only reservation reform by the street protest they would provoke; separate what data identify from what is assumed.
Entities	Agents (120,000, each about 12,200 people) with social group, SC/ST tier, income position, eligibility segment, identity strength and threshold; neighbourhoods of about 100 agents; districts and states (grounded); organizations and government as rules, not agents.
Scales	One step is a day; campaigns last 40 days; space is the 640 Census 2011 districts (grounded) or national (stylized).
Process overview	Each day: grievance, social pull and organizational push determine each agent's protest probability; decisions; fatigue; turnout updates; bandh calls and concession (grounded); deaths and the martyr effect (Section 6).
Design concepts	Emergence: turnout, bandh timing, concession, deaths. Adaptation: none (thresholds rise with fatigue). Interaction: neighbourhood and national social influence. Stochasticity: decisions, deaths, organization, world draws; paired seeds across scenarios. Observation: peak-day turnout, unique participants, protester-days, deaths, state and group shares, event counts through the observation model.
Initialization	Population from Census and NFHS shares with seed 7; parameters from the specification and, per run, a world draw (grounded: one retained calibration set).
Input data	Rank lists and allotments (allocation model), Census 2011 and NFHS-5 (population), GDELT event counts and reported deaths (calibration); none enter during a run.
Submodels	Grievance (Eq. 1), social pull (Eq. 2), decision (Eq. 3), bandh calls, concession and backlash, deaths; parameters in Table 4.

G Stylized specification: channel decomposition

The same decomposition in the stylized specification, 50 paired runs: L5's party-amplifier channel alone changes the peak by -39% and its symbolic channel alone by -60% ; L3's symbolic channel gives -91% and its material channel -44% ; L1's material channel gives -57% and its symbolic channel -17% . Heavy policing's turnout cost alone gives -51% and its extra violence and martyr response alone $+42\%$; the full scenario multiplies deaths by 3.9, of which the assumed multiplier of 3 is input and a factor of 1.3 comes from the change in protester-days ($+20\%$).

H Grounded specification: Morris screening

Table 25: Morris screening in the grounded specification (20 trajectories around the central calibrated values): μ^* on the baseline log peak and on the paired peak change of L1, L3 and L5, and σ for the log peak.

Parameter	Log peak	L1	L3	L5	σ (log peak)
Symbolic weight w_s	3.00	51	8	46	1.54
Mean threshold $\bar{\theta}$	1.57	30	8	27	1.15
Threshold spread σ_θ	0.94	23	15	21	1.19
Material weight w_m	0.77	14	2	13	0.80
Deprived-tier threat share	0.57	19	2	18	0.64
Loss aversion λ	0.51	12	2	15	0.51
Decision noise τ	0.45	9	5	11	0.56
Party amplifier a	0.31	9	3	6	0.21
Bandh mobilization b	0.21	4	4	2	0.18
Neighbourhood saturation n^*	0.19	6	1	7	0.22
Martyr effect δ	0.17	8	1	7	0.17
National influence β_v	0.14	2	1	4	0.07
Neighbourhood influence β_n	0.14	2	1	2	0.13
Death rate κ	0.13	6	2	10	0.19
Fatigue f	0.08	3	0	5	0.13

I Code map

File	Contents
merit_allocation/	Rank-list data, latent merit fit, over-and-above and reserved-first choice rules, merged-pool regimes
data_pipelines/	Rank-list extraction, GDELT download and filtering, episode definitions, calibration targets
protest_simulation/synthetic_population.py	National population, tiers, income line, eligibility flags, neighbourhoods
protest_simulation/geography.py	District population from Census 2011 and NFHS-5
protest_simulation/model_parameters.py	Parameters and structural options
protest_simulation/protest_campaign.py	Daily dynamics
protest_simulation/policy_interventions.py	Scenario catalogue at reference values
protest_simulation/intervention_priors.py	Priors over intervention effects
protest_simulation/episode_simulation.py	Historical episodes as model shocks
protest_simulation/grounded_specification.py	Grounded specification
protest_simulation/monte_carlo.py, parallel.py	Paired runs, bootstrap intervals, multiprocessing
experiments/	One script per analysis in this paper
tests/	Regression, property and reimplementations tests